High Spatial Resolution Soil Moisture Mapping Using a Lobe Differencing Correlation Radiometer on a Small Unmanned Aerial System

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***Abstract*— A persistent challenge in the measurement of soil moisture from satellites stems from the inherently low spatial resolution using active or passive system. The lobe differencing correlation radiometer (LDCR) on a small unmanned aerial system (sUAS) is shown to provide a capability to measure soil moisture at high spatial resolution for a range of scientific and operational purposes. Flight tests of LDCR on a fixed wing sUAS were performed at the Canton, Oklahoma Soilscape site in September 2015, and Irrigation Research Foundation (IRF) in Yuma, Colorado, in June 2016. The LDCR design and performance are discussed, and the calibration using both preflight lab test data and in-flight data over a calm pond was performed to calibrate the radiometer. Radio frequency interference (RFI) from the sUAS platform was observed and mitigated. The LDCR sampling processes are detailed and an implementation of the $\tau - \omega$ vegetation correction model along with a semiempirical surface roughness correction model incorporating a full-domain soil moisture mapping algorithm is presented. The algorithm uses a weakly nonlinear observation operator suitable for irregular sUAS flight trajectories that maps volumetric soil moisture (VSM) on a user-defined product grid from the sUAS sampling grid. Using LDCR radiometric and thermal measurements, along with Landsat-based vegetation water content (VWC) and soil texture information, soil moisture was mapped at decameter spatial resolution. The retrieved VSM data are favorably compared with *in situ* VSM measurements and irrigation records. A method for determining LDCR VSM estimation errors is developed to quantify the mapping algorithm accuracy and assess the impact of error sources.**

***Index Terms*— High spatial resolution soil moisture map, linear minimum mean square error (LMMSE), radio frequency interference (RFI) mitigation, small unmanned aerial system (sUAS), soil moisture retrieval algorithm, soil-vegetation radiative transfer (SVRT) model.**

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I. Introduction

SOIL moisture that is of fundamental importance to many hydrological, biological, and geochemical processes plays an important role in the development and evolution of convective weather and precipitation, and impacts water resource management, agriculture, and flood runoff prediction. Passive microwave remote sensing at L-band (1–2 GHz) is one of the most promising approaches to operationally measure soil moisture. ESA's Soil Moisture and Ocean Salinity (SMOS) mission launched in November 2009 and NASA's Soil Moisture Active Passive (SMAP) mission launched in January 2015 have provided spaceborne global measurements of soil moisture and surface freeze–thaw state at fixed crossing times and with spatial resolutions as low as 9 km for some products [1]–[4]. However, the observation of soil moisture on much smaller spatial scales and at arbitrary diurnal times for satellite data validation, precision agriculture, and evaporation and transpiration studies of boundary layer heat transport is yet needed.

The first retrieved soil moisture maps using a unmanned aerial vehicle (UAV)-borne L-band radiometer in 2010 demonstrated the capabilities of drone platforms for high spatial resolution soil moisture mapping [34]. Such capabilities have recently been further demonstrated using a unique lobe differencing correlation radiometer (LDCR) on a small unmanned aerial system (sUAS) platform and are reported herein. The LDCR was developed by the University of Colorado's Center for Environmental Technology (CET) and Black Swift Technology, LLC (BST) as a new means of airborne mapping of soil moisture on spatial scales as small as several meters (i.e., approximately the height of the aircraft platform). Compared with various other methods of soil moisture measurement based on either *in situ* measurements [5], [6] or existing airborne sensors suitable for manned aircraft deployment [7], [8], the integrated design of the LDCR on a lightweight sUAS provides several distinct advantages, foremost of which is subwatershed (∼km scale) coverage at very high (∼decameter) spatial resolution. Data from the LDCR system is suitable for scaling studies, watershed and cropland management, and other applications such as trafficability at comparatively low fixed and operating costs.

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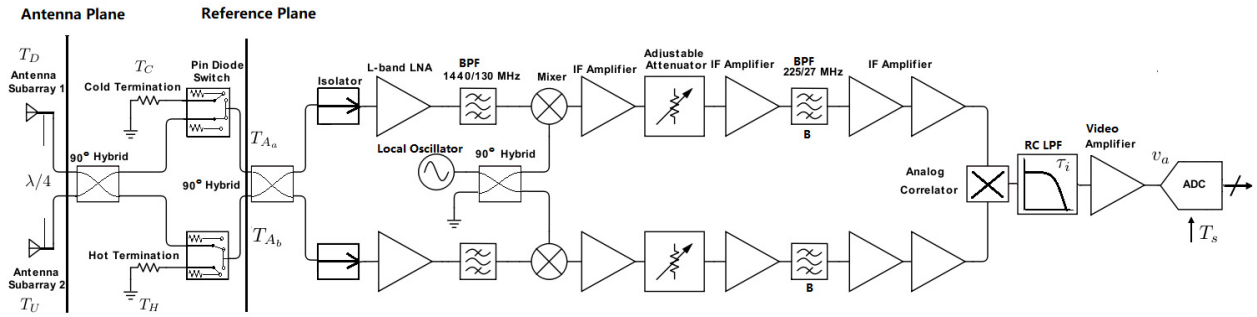

Fig. 1. LDCR Rev A block diagram, including antenna and calibration reference planes.

L-band observations require radiometric absolute accuracy of order ∼1 K or better in the upwelling antenna temperature measurements [9]. This level of accuracy was achieved using the LDCR by differencing the upwelling antenna temperature with the well-known downwelling temperature of cold space. Accordingly, the LDCR antenna was designed to have lobes that independently view both nadir and zenith using a unique lightweight microstrip collinear (MiCo) structure [9]. The LDCR measures the difference in antenna temperature between these two lobes.

To support the development of the LDCR system, flight tests using LDCR and the BST Tempest sUAS were performed at the Canton, Oklahoma Soilscape [6], [50] site, in September 2015 and Irrigation Research Foundation (IRF) in Yuma, Colorado, in June 2016. The LDCR antenna temperature T_A and surface physical temperature T_P were measured as primary inputs to map retrieved soil moisture over regions of size of approximately one-quarter of a section. However, there are several challenging aspects to the use of sUAS L-band data for retrieving volumetric soil moisture (VSM), including: 1) sUAS flight lines are inherently irregular due to wind variations, line of site flight, and obstacle avoidance requirements, and other operational considerations; 2) the desired regular mapping grid is more dense than typical flight patterns will permit; and 3) the vegetation cover is often inhomogeneous at the spatial scale of the available observations. To accommodate these unique spatial attributes, a new high spatial resolution soil moisture mapping algorithm was developed specifically for sparse irregular sUAS observations. The LDCR calibration method, LDCR VSM retrieval algorithm, field campaigns, T_P and VSM maps, and intercomparisons with *in situ* soil moisture data and irrigation records are detailed herein.

II. LDCR Hardware

The first LDCR revision (denoted as Rev A) had been built and integrated into the Black Swift Technology LLC (BST) Tempest sUAS aircraft in 2015. The LDCR RF frontend uses two identical receiving channels with a p-i-n diode switch in each channel for calibration purposes prior to analog correlation (Fig. 1). The switches enable either the antenna ports or hot and cold internal calibration sources to be periodically connected to each quadrature-summed radiometer channel. As a result, the LDCR can be operated in one of four switched calibration states. When the antenna ports are connected, the quadrature phase difference provided by the first 90° coupler, along with the 90° difference provided by the MiCo antenna's quarter-wavelength vertical separation [10], effectively implements upwelling and downwelling antenna lobes. The associated upwelling and downwelling signals are defined at the reference plane shown in Fig. 1. This lobe differential architecture using the stable downwelling brightness temperature of cold space as a reference, meanwhile the internal calibration references are used to determine the voltage offset and thermal drift of the whole system.

In the following discussion, we consider the analog data observed in each of the four switch states. In each of these states, the analog correlator output voltage is linearly related to the temperature differences at the reference plane. The LDCR sensitivity and offset voltage are determined by preflight laboratory characterization, then validated in flight using four switch state data. Physical temperature effects that induce receiver drift are accommodated in the calibration process.

A. LDCR Calibration Swtich States

The analog correlator output voltage v_a after *RC* low pass filter and video amplifier for all switch states can be expressed as

$$\langle v_a \rangle = G_o \mathrm{kB}(T_{A_b} - T_{A_a}) + v_o \tag{1}$$

where T_{A_a} and T_{A_b} are input antenna temperature for channel a and channel b as defined at the reference plane for each of four switch states, and $\langle \cdot \rangle$ denotes time averaging. The offset voltage v_o is the output given a null antenna temperature difference input, and k is the Boltzmann's constant. The gain G_o (in mV/W) is proportional to the geometric mean of the power gain of the two LDCR amplifying channels and *RC* low pass filter integration time constant τ_i. The instrument bandwidth B is 27 MHz (i.e., from 1400 to 1427 MHz) and $\tau_i = RC$ is 1.5 ms.

The LDCR gain G_o is approximately estimated based on the sum of the gains and losses of all components including the RF, downconversion, and IF components, analog correlator, and video amplifier. Assuming the first 90° hybrid coupler is balanced and the two receiving channels are uncoupled, the temperatures T_{A_a} and T_{A_b} at the reference

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plane are

$$T_{A_a} = \begin{cases} \alpha_{a_c} T_C + (1-\alpha_{a_c}) T_{P_{a_c}}, & \text{internal source} \\ \alpha_{a_s} T_D + (1-\alpha_{a_s}) T_{P_{a_s}}, & \text{antenna input} \end{cases} \quad (2)$$

$$T_{A_b} = \begin{cases} \alpha_{b_c} T_H + (1-\alpha_{b_c}) T_{P_{b_c}}, & \text{internal source} \\ \alpha_{b_s} T_U + (1-\alpha_{b_s}) T_{P_{b_s}}, & \text{antenna input} \end{cases} \quad (3)$$

where T_C and T_H are the cold and hot internal calibration source physical temperatures, and T_U and T_D are the LDCR observed upwelling and downwelling antenna temperatures which are detailed in (21). The internal calibration temperatures are typically cooled to ~273 K for T_C and heated to ~328 K for T_H. The correction factor α_{a_c} is proportional to the internal calibration source path RF loss from the internal source to the reference plane, and the correction factor α_{a_s} is proportional to the antenna input path RF loss from the antenna plane to the reference plane for channel a (Fig. 1). Similarly, α_{b_c} and α_{b_s} are correction factors for channel b. The internal source paths and antenna input path have physical temperatures of $T_{P_{a_c}}$, $T_{P_{a_s}}$, $T_{P_{b_c}}$, and $T_{P_{a_s}}$ (respectively).

On the LDCR receiving board, the microstrip transmission lines to the left of the reference plane in both channels are fabricated to be mirror-symmetric. For the p-i-n diode switches used (Hittite HMC284MS8G) the insertion loss for either path is the same to within 0.1 dB. Accordingly, the correction factors in channel a and b are assumed to have equal values to within a few percent, that is $\alpha_c = \alpha_{a_c} \cong \alpha_{b_c}$ and $\alpha_s = \alpha_{a_s} \cong \alpha_{b_s}$. The physical temperatures of all paths have the same value T_{P_i} to within 0.1 K. With these assumptions, the reference temperature differences are modeled as

$$\begin{aligned} &\text{Internal Souce State: } T_{A_b} - T_{A_a} = \alpha_c (T_H - T_C) \\ &\text{Scene State: } \quad T_{A_b} - T_{A_a} = \alpha_s (T_U - T_D) \\ &\text{Mixed Hot-Downwelling State:} \\ &T_{A_b} - T_{A_a} = \alpha_c T_H - \alpha_s T_D + (\alpha_s - \alpha_c) T_{P_i} \\ &\text{Mixed Upwelling-Cold State:} \\ &T_{A_b} - T_{A_a} = \alpha_s T_U - \alpha_c T_C + (\alpha_c - \alpha_s) T_{P_i} \end{aligned} \quad (4)$$

to within a temperature bias of less than ~3–4 K in the worst case of parameter error. The internal calibration source correction factor α_c is estimated from the internal calibration source emissivity (> 99%) and p-i-n diode switch insertion loss (0.7 dB) to be $10^{-0.7/10} \times 99\% = 0.84$. The value of α_s for the antenna input is similarly estimated using various RF losses, including the coaxial and microstrip line loss (1.2 dB), the insertion loss of the first 90° hybrid coupler (0.25 dB), and the p-i-n diode switch insertion loss (0.7 dB). The value of α_s is thus approximately $10^{-(1.2+0.25+0.7)/10} = 0.61$.

For the four switch states, the analog output voltage can be expressed as

$$\begin{aligned} &\text{Internal Souce State: } \langle v_a \rangle_{HC} = \alpha_c G_o k B (T_H - T_C) + v_o \\ &\text{Scene State: } \quad \langle v_a \rangle_{UD} = \alpha_s G_o k B (T_U - T_D) + v_o \\ &\text{Mixed Hot-Downwelling state:} \\ &\langle v_a \rangle_{HD} = G_o k B (\alpha_c T_H - \alpha_s T_D + (\alpha_s - \alpha_c) T_{P_i}) + v_o \\ &\text{Mixed Upwelling-Cold State:} \\ &\langle v_a \rangle_{UC} = G_o k B (\alpha_s T_U - \alpha_c T_C + (\alpha_c - \alpha_s) T_{P_i}) + v_o. \end{aligned} \quad (5)$$

The calibration process is based on internal source and scene switch states with the measurements from two mixed switch states used as supplementary data for bias correction purposes.

The LDCR ΔT_{rms} is a key performance statistic describing the minimum 1-sigma detectable temperature change. Defined with respect to the reference plane, it is

$$\begin{aligned} \Delta T_{\text{rms}} &= \frac{\sigma_{v_a}}{\dfrac{\partial \langle v_a \rangle}{\partial (T_{A_b} - T_{A_a})}} \\ &= \frac{\sqrt{T_{A_a}^2 + T_{A_b}^2 + (T_{R_a} + T_{R_b})(T_{A_a} + T_{A_b}) + 2 T_{R_a} T_{R_b}}}{\sqrt{B \tau_i}} \end{aligned} \quad (6)$$

where σ_{v_a} is the correlator output voltage standard deviation and T_{R_a} and T_{R_b} are the receiver noise temperatures of the two LDCR receiving channels. Accordingly, $\sigma_{v_a} = G_o k B \Delta T_{\text{rms}}$. Using these relations, the values of T_{R_a} and T_{R_b} were empirically determined by parameter fitting to be 500 $\pm$10 K for LDCR Rev A by measuring σ_{v_a} using ~2 $\times 10^6$ independent samples and presenting a variety of known input antenna temperatures T_{A_a} and T_{A_b}. In this process, it is presumed that $T_{R_a} = T_{R_b}$ and that G_o, B, τ_i, α_c, and α_s are accurately known based on estimation. For the internal source and scene switch state, the output voltage and temperature differences of $T_H - T_C$ and $T_U - T_D$ are related to $T_{A_b} - T_{A_a}$ by (4). Consequently, the ΔT_{rms} values for these two switch states are modified as

$$\begin{aligned} \text{Internal Souce State: } \Delta T_{\text{rms},c} &= \frac{\sigma_{v_a}}{\dfrac{\partial \langle v_a \rangle}{\partial (T_H - T_C)}} = \frac{\Delta T_{\text{rms}}}{\alpha_c} \\ \text{Scene State: } \Delta T_{\text{rms},s} &= \frac{\sigma_{v_a}}{\dfrac{\partial \langle v_a \rangle}{\partial (T_U - T_D)}} = \frac{\Delta T_{\text{rms}}}{\alpha_s} \end{aligned} \quad (7)$$

where the modified values refer to fundamental precisions possible for measuring $T_H - T_C$ and $T_U - T_D$.

B. Preflight Calibration

The LDCR preflight calibration process determines the offset voltage v_o and sensitivities $\alpha_c G_o$kB and $\alpha_s G_o$kB (in mV/K), all of which are found to be primarily dependent on the physical temperature of the analog correlator chip. The offset voltage v_o is independent of switch states, while the sensitivities $\alpha_c G_o$kB and $\alpha_s G_o$kB differ by 0.4–0.7 dB. The offset v_o was estimated using equal physical temperatures at the reference plane for the two channels, and the sensitivities for the internal source and scene states were estimated using known antenna temperature inputs as follows.

The sensitivity $\alpha_s G_o$kB for the scene state was estimated by connecting a matched load to each of the two LDCR inputs through a connectorized 90° hybrid coupler. By this method the dual-lobe characteristics of the antenna were simulated. One of the matched loads was placed in liquid nitrogen (77.3 K), and the other left at room temperature (~295 K). The feedline and quadrature hybrid losses were properly accounted

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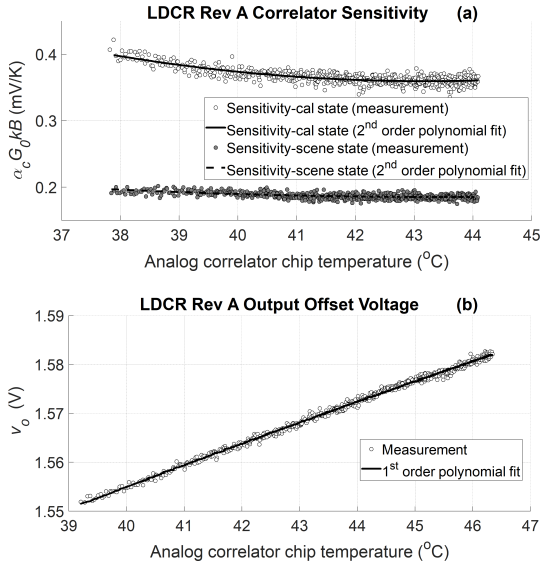

Fig. 2. LDCR Rev A (a) sensitivities and (b) offset voltages as functions of analog correlator temperature.

for in this process to provide a precision in $T_U - T_D$ of ~2.5 K. The product $\alpha_c G_o$kB for the internal source state was similarly estimated using the internal calibration sources at known measured temperatures T_H and T_C to a precision in $T_H - T_C$ of ~0.5 K.

The estimated sensitivities and offset voltage as functions of analog correlator chip physical temperature are shown in Fig. 2. It is noted that: 1) the sensitivities decrease appreciably with correlator chip temperature; and 2) the offset voltage increases approximately linearly with correlator chip temperature from ambient to 319 K. Polynomial fits were determined and subsequently used as models for these quantities. Based on the estimated sensitivities and offset voltages and their estimation precisions, the LDCR upwelling antenna temperature T_U was determinable by the above preflight procedure to a precision estimated to be ~5–6 K in bias.

C. In-Flight Calibration

The preflight calculated sensitivities were validated using in-flight data observed when the sUAS flew over a calm fresh water pond for ~7 s at ~15 m altitude. The measured upwelling and downwelling antenna temperatures T_U and T_D were calculated from the measured pond surface temperature T_{P_w} of 298 K and the well-known L-band downwelling sky brightness temperature of 5 ± 0.4 K, which is the cold space temperature compensated by ~0.025 dB of atmospheric loss [11]. The upwelling T_U calculation used the measured polarimetic LDCR antenna gain pattern integrated over the upwelling terrain of view which includes pond and surrounding field, and the Fresnel reflectivity coefficient for flat fresh water. The measured antenna and radome ohmic efficiency of $\eta_l = 78 \pm 5\%$ [10] and measured antenna physical temperature T_{P_A} were also considered in this calculation (see §IV). The water and terrain physical temperature was measured by the LDCR IR sensor where the terrain had a physical temperature to be ~8 K higher than that of the water. The contribution of the land surrounding the pond to the overall T_U calculation for the selected pond flight line is 0.2–1.5 K.

From (5), the LDCR analog output voltage vector (minus offset v_o) for all switch states can be expressed as

$$\begin{bmatrix} \langle v_a \rangle_{UD} - v_o \\ \langle v_a \rangle_{UC} - v_o \\ \langle v_a \rangle_{HD} - v_o \\ \langle v_a \rangle_{HC} - v_o \end{bmatrix} = \begin{bmatrix} T_U - T_D & 0 \\ T_U - T_{P_i} & T_{P_i} - T_C \\ T_{P_i} - T_D & T_H - T_{P_i} \\ 0 & T_H - T_C \end{bmatrix} \begin{bmatrix} \alpha_s G_o kB \\ \alpha_c G_o kB \end{bmatrix} + \overline{n} = \overline{\overline{W}}_c \overline{x} + \overline{n} \quad (8)$$

where $\overline{\overline{W}}_c$ is the calibration observation operator, $\overline{x}$ is the sensitivity vector to be determined, and $\overline{n}$ is the observation noise vector. The sensitivities were subsequently determined by unbiased linear minimum mean square error (LMMSE) estimation using [12]

$$\begin{bmatrix} \widehat{\alpha}_s G_o kB \\ \widehat{\alpha}_c G_o kB \end{bmatrix} = \overline{\overline{D}}_c \begin{bmatrix} \langle v_a \rangle_{UD} - v_o \\ \langle v_a \rangle_{UC} - v_o \\ \langle v_a \rangle_{HD} - v_o \\ \langle v_a \rangle_{HC} - v_o \end{bmatrix} - \overline{\overline{D}}_c \ \overline{\overline{W}}_c \begin{bmatrix} \langle \alpha_s \rangle G_o kB \\ \langle \alpha_c \rangle G_o kB \end{bmatrix} + \begin{bmatrix} \langle \alpha_s \rangle G_o kB \\ \langle \alpha_c \rangle G_o kB \end{bmatrix} \quad (9)$$

where are $\langle \alpha_s \rangle$ and $\langle \alpha_c \rangle$ are preflight determined sensitivities. The LMMSE solution provides the calibration operator $\overline{\overline{D}}_c$ which minimizes the sensitivities estimation error $\overline{e} = \widehat{\overline{x}} - \overline{x}$. For physical LMMSE estimation, $\overline{\overline{D}}_c$ is calculated as [12]

$$\overline{\overline{D}}_c = \overline{\overline{R}}_{xx} \overline{\overline{W}}_c^t \left(\overline{\overline{W}}_c \ \overline{\overline{R}}_{xx} \overline{\overline{W}}_c + \overline{\overline{R}}_{nn} \right)^{-1} \quad (10)$$

where $\overline{\overline{R}}_{xx}$ and $\overline{\overline{R}}_{nn}$ are the sensitivity and observation noise covariance matrices. The sensitivity variations about the preflight values and observation noises for different switch states are considered to be uncorrelated. The observation noise was calculated from the LDCR analog output voltage standard deviation to be 2.8 mV, and the sensitivity error variance was estimated to be ~10% based on the available precision in determining the internal source and antenna input path RF losses.

The in-flight sensitivities were estimated using (9) to be 0.34 and 0.19 mV/K for the internal source and scene state (respectively). The estimation error $\overline{\overline{R}}_{ee}$ in this process can be calculated as

$$\overline{\overline{R}}_{ee} = \left(\overline{\overline{I}} - \overline{\overline{D}}_c \ \overline{\overline{W}} \right) \overline{\overline{R}}_{xx} \left(\overline{\overline{I}} - \overline{\overline{D}}_c \ \overline{\overline{W}} \right)^t + \overline{\overline{D}}_c \ \overline{\overline{R}}_{nn} \overline{\overline{D}}_c^t \quad (11)$$

where $\overline{\overline{I}}$ is the diagonal matrix. The sensitivity estimation errors for both internal source and scene state were thus calculated to be 2%. The preflight sensitivities were 0.36 and 0.19 mV/K for the same pond overflight instrument temperature of 316 K. Although the in-flight and lab-based sensitivities are similar to each other, we use the in-flight values to correct the fitted models in Fig. 2 for all in-flight calibration purposes.

The *a priori* values of sensitivities $\alpha_c G_{\text{okB}}$, $\alpha_s G_{\text{okB}}$, analog output voltage standard deviation σ_{v_a}, and ΔT_{rms} values are calculated using (6) and (7), and summarized in Table I along with the preflight and in-flight estimated values. The observed sensitivities are lower by 0.1–1.2 dB than *a priori* values

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TABLE I
LDCR SYSTEM PARAMETERS FOR $\tau_i = 1.5$ MS TIME CONSTANT AND $T = 1$ MS SAMPLING TIME INTERVAL, INCLUDING QUANTIZATION ERROR

LDCR Rev A (analog correlator chip physical temperature of 316 K)						
	Internal Source state			Scene state		
Parameter	Pre-flight	In-flight	A-priori	Pre-flight	In-flight	A-priori
$\alpha_c G_o kB$ $\alpha_s G_o kB$ (mV/K)	0.36	0.34	0.39	0.19	0.19	0.28
σ_{v_a} (mV)	2.8	2.8	2.8	2.8	2.8	2.8
$\Delta T_{RMS,c}$ $\Delta T_{RMS,s}$ (K)	7.77	8.25	7.18	14.74	14.74	9.29

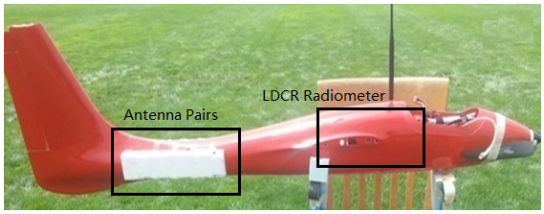

Fig. 3. LDCR Rev A integration into the Tempest sUAS.

calculated based on estimated values of G_o, α_c and α_s. The measured and *a priori* values of σ_{v_a} are approximately the same (Table I).

The sampling interval T_s of the analog correlator output voltage for both LDCR revisions is 1 ms. Given a typical ∼20 m/s sUAS air speed and decameter (∼50 m) antenna 3-dB foot print size, it is reasonable to average the sampled data over ∼0.5 second of flight. The output voltage standard deviation σ'_{v_a} after averaging can be calculated as

$$\sigma'_{v_a} = \frac{\sigma_{v_a}}{\sqrt{N_a}} \left[\frac{N_a(1-\rho^2) - 2\rho(1-\rho^{N_a})}{N_a(1-\rho)^2} \right]^{1/2} \tag{12}$$

where $\rho = e^{-(T_s/\tau_i)}$ is the intersample correlation cause by *RC* low pass filtering, and N_a is the number of samples being digitally averaged. The integration time was allocated equally to the four switch states in Rev A with each calibration cycle lasting 42 ms. Thusly, N_a is ∼100 and the pixel-averaged $\Delta T_{\mathrm{rms},s}$ for the scene state is reduced to 2.58 K for Rev A.

D. LDCR Integration Into sUAS and RFI Mitigation

Radio frequency interference (RFI) from the sUAS is one of the main concerns in their use for passive remote sensing. Upon integration of the LDCR Rev A into the Tempest sUAS (Fig. 3), the motor and motor controller were found to radiate wide band noise which coupled through LDCR MiCo antennas. This radiation cause the antenna temperature noise to increase to up to ∼4000 K in one local flight test performed in August 2015 (Fig. 4). After shielding all potential radiating sources, including the motor, motor controller, battery, and all wire connections, the noise was decreased to a level less than the output voltage standard deviation (σ_{v_a}) for the all Tempest follow-up flights. Interference from the microcomputer was not detected.

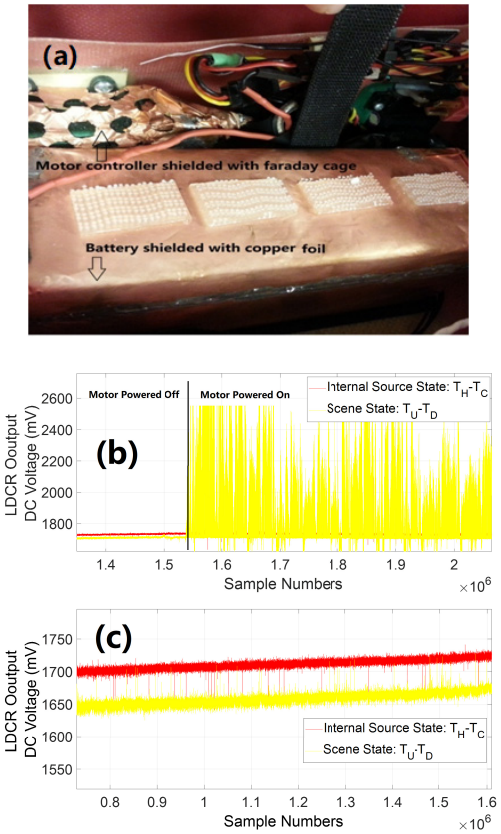

Fig. 4. Tempest sUAS and LDCR Rev A (a) shielding techniques, and analog output voltage noise level, (b) before and (c) after shielding.

III. SOIL MOISTURE RETRIEVAL BACKGROUND

The LDCR mapping algorithm builds upon soil moisture retrieval techniques established for the SMOS and SMAP missions. However, due to the unique LDCR polarization-mixed signal, broad antenna beamwidth, and irregular sampling pattern, the algorithm necessarily departs from that of space borne missions. The operational SMOS retrieval algorithm is based on the multiangular and dual-polarization observing capabilities of the SMOS interferometric sensor. Soil moisture and vegetation opacity are simultaneously retrieved using the SMOS multiangle polarimetric brightness temperature imagery [14]. The operational soil moisture retrieval method for SMAP uses a single vertically polarized channel in the current baseline algorithm [15]. The vegetation opacity is not simultaneously retrieved. Instead, it is calculated from vegetation water content (VWC) measurements which are estimated using normalized differential vegetation index (NDVI) data from NASA's Moderate Resolution Imaging Spectroradiometer (MODIS).

For LDCR, antenna temperature samples of mixed polarization and multiple incidence angles are directly measured, and the vegetation opacity is estimated using VWC values from temporally interpolated Landsat normalized differential water index (NDWI) data. An observation operator using VSM data defined on a regular user-defined product grid and computing the observed T_A on an irregular sUAS flight sampling grid was developed to support this method. The integral

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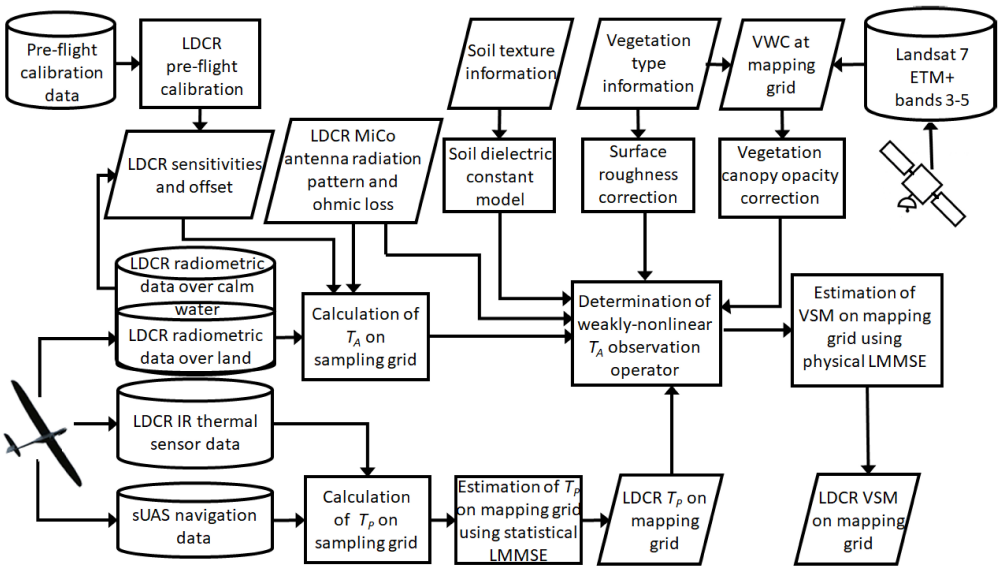

Fig. 5. LDCR VSM retrieval and mapping algorithm.

expression for the LDCR T_A samples is based on the $\tau - \omega$ vegetation opacity model [9], [16]–[19] incorporating VWC correction, an empirical soil dielectric mixing model [20], [21], and surface roughness correction provided by the H-Q-N model [14]. A linear fit to the integral expression is used to approximate the weakly nonlinear observation operator.

The LDCR high spatial resolution T_A data are subsequently mapped to VSM on a user-defined grid using a LMMSE estimation method (Fig. 5). The LMMSE inversion scheme has been widely applied as an optimal estimator in passive and active remote sensing for decades [22]. The mixture of polarizations and incidence angles is addressed in this retrieval formulation along with the spatial covariance of VSM as relevant for managed cropland of various vegetation types and resulting VSM error variance. *A priori* information on vegetation and surface homogeneity is shown to be readily imposed on the retrieval operator by presetting the off-diagonal elements in the VSM spatial covariance matrix.

IV. Soil-Vegetation Radiative Transfer Model

For passive remote sensing of soil moisture, a soil-vegetation radiative transfer (SVRT) solution incorporating scattering to zeroth order is commonly used for tenuous vegetation cover such as cultivated crops and grasses [23], and to first order for heavy vegetation covers such as shrubs and trees [24]. In this study, we focus on soil moisture mapping over mostly cultivar and grass vegetation cover. Accordingly, the $\tau-\omega$ zeroth order RT model is used for canopy attenuation.

A. $\tau - \omega$ Model

In the $\tau-\omega$ model, the vegetation optical opacity $\tau_{c_p}(\theta)$ and effective canopy scattering albedo $\omega_p(\theta)$ are used to parametrize radiative transmission, where p is either vertical (v) or horizontal (h) polarization, and θ is the incidence angle from nadir [14], [25]. The thermal emission from a scene consisting

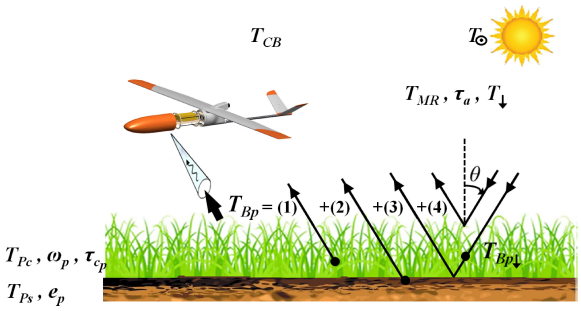

Fig. 6. Radiation streams in the $\tau - \omega$ SVRT model.

of a vegetation layer above an infinite half-space representing the subjacent soil medium can be written as the sum of four terms: 1) the upwelling vegetation emission; 2) upwelling soil emission attenuated by the vegetation canopy; 3) the downwelling canopy and sky emission reflected by the soil and attenuated by the cannopy layer; and 4) the downwelling sky emission reflected by the canopy layer (Fig. 6). The brightness temperature for polarization p at the canopy top can be expressed as

$$T_{B_p} = (1-\omega_p)(1-e^{-\tau_{c_p}})T_{P_c} + e_p e^{-\tau_{c_p}} T_{P_s} + (1-e_p)e^{-\tau_{c_p}} T_{B_p\downarrow} + \omega_p T_{\downarrow} \quad (13)$$

where e_p is the soil emissivity at the presumed smooth soil–air interface for polarization p, and T_{P_c} and T_{P_s} are canopy and soil surface physical temperatures. The temperature $T_{\downarrow}$ is the downwelling sky brightness temperature calculated from the cold space brightness temperature T_{CB} of 2.73 K with atmospheric attenuation compensation [26]

$$T_{\downarrow}(\theta) = (1-e^{-\tau_a \sec\theta})T_{MR} + T_{CB}e^{-\tau_a \sec\theta} \quad (14)$$

where τ_a is the atmospheric opacity at L-band, and T_{MR} is the atmospheric mean radiating temperature [26].

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TABLE II
LAND COVER PARAMETERS FOR SVRT VEGETATION CORRECTION AND SURFACE ROUGHNESS CORRECTION

Land cover/vegetation type	ω_p	b_h	b_v	h_R
Tall grass (Canton)	0.05	0.57	0.43	0.156
Short grass (IRF)	0.05	0.13	0.13	0.156
Corn	0.10	0.18	0.20	0.094
Wheat & grain sorghum	0.00	0.10	0.30	0.083
Soybean & sugar beets	0.00	0.20	0.20	0.148
Forest	0.05	0.10	0.10	0.160
Built	0.4-0.95	∞	∞	0.000
Bare Soil (road ways)	0.00	0.00	0.00	0.150

The downwelling brightness temperature at the soil surface $T_{Bp\downarrow}$ is expressed as

$$T_{B_p\downarrow} = (1-\omega_p)(1-e^{-\tau_{c_p}})T_{P_c} + (1-\omega_p)e^{-\tau_{c_p}}T_{\downarrow}. \quad (15)$$

The canopy scattering albedo ω_p is found to be dependent on vegetation type [14], [16], [17]. The vegetation opacity τ_{c_p} can be expressed as

$$\tau_{c_p}(\theta) = b_p W_v \sec\theta \quad (16)$$

where W_v is the VWC which can be calculated from vegetation indexes, such as NDVI or NDWI data, and b_p is an empirical coefficient dependent on vegetation type and polarization [27]. The parameters for vegetation correction are listed in Table II [15], [27].

It is recognized that soil surface and vegetation canopy physical temperatures might differ by up to ~2–3 K [14]. In addition, the soil physical temperature over the L-band probing depth of ~1–~10 cm might have a difference of several K [14]. In this study, the soil and vegetation canopy are assumed to have the same physical temperature, that is $T_{P_c} = T_{P_s} = T_P$. Given this assumption, the T_{B_p} can be approximated as

$$\begin{aligned} T_{B_p} &= (1-\omega_p)(1-e^{-\tau_{c_p}})T_P[1+(1-e_p)e^{-\tau_{c_p}}] \\ &\quad + e_p e^{-\tau_{c_p}}T_P + (1-\omega_p)(1-e^{-\tau_{c_p}})e^{-2\tau_{c_p}}T_{\downarrow} + \omega_p T_{\downarrow}. \end{aligned} \quad (17)$$

Specular soil emissivity can be calculated from soil reflectivity r_p as $e_p = 1 - r_p$. For a smooth soil surface, the Fresnel reflectivities at the soil-air interface can be expressed as

$$r_v(\theta) = \left|\frac{\epsilon_s\cos\theta - \sqrt{\epsilon_s - \sin^2\theta}}{\epsilon_s\cos\theta + \sqrt{\epsilon_s - \sin^2\theta}}\right|^2 \quad (18)$$

$$r_h(\theta) = \left|\frac{\cos\theta - \sqrt{\epsilon_s - \sin^2\theta}}{\cos\theta + \sqrt{\epsilon_s - \sin^2\theta}}\right|^2. \quad (19)$$

These specular reflectivities are later modified by surface roughness corrections factors.

The LDCR microstrip collinear (MiCo) antenna synthesizes two broad main beams pointing primarily to the nadir and zenith directions along with minor back lobes pointing in the opposite directions [10]. The LDCR thus observes upwelling and downwelling antenna temperatures T_U and T_D defined (respectively) as the brightness temperatures integrated over these two beams

$$\begin{aligned} T_{U/D} &= \eta_l \frac{1}{4\pi}\iint_{2\pi}[G_{vv,U/D}T_{B_v} + G_{hh,U/D}T_{B_h} \\ &\quad + \mathrm{Re}\{G_{vh,U/D}\}T_{B_U} + \mathrm{Im}\{G_{hv,U/D}\}T_{B_V}]d\Omega \\ &\quad + \eta_l \frac{1}{4\pi}\iint_{2\pi}[G_{vv,U/D} + G_{hh,U/D} \\ &\quad + \mathrm{Re}\{G_{vh,U/D}\} + \mathrm{Im}\{G_{hv,U/D}\}] \\ &\quad \times T_{\downarrow}d\Omega + (1-\eta_l)T_{P_A} \end{aligned} \quad (20)$$

where T_{B_U} and T_{B_V} are the upwelling third and fourth Stokes brightness temperature parameters, $G_{vv,U/D}$, $G_{hh,U/D}$, $G_{vh,U/D}$, and $G_{hv,U/D}$ are the MiCo antenna gain matrix elements for the main beam pointing to the nadir/zenith directions (respectively), and T_{P_A} is the physical temperature of the antenna pairs measured by thermistors during flight. The thin layer (~50 m) of atmosphere between the canopy top and sUAS is neglected. Therefore, the downwelling sky brightness temperature defined at canopy top $T_{\downarrow}$ is used for the integration over the upper hemisphere in (8). In addition, the polarization effects on the downwelling sky brightness temperature $T_{\downarrow}$ are negligible at L-band for this application [11], [13].

Based on both simulation and measurement of LDCR MiCo antenna radiation patterns, the $G_{vh,U/D}$ and $G_{hv,U/D}$ terms are smaller in magnitude than $G_{vv,U/D}$ and $G_{hh,U/D}$ by over 20 dB [10]. Additionally, the third and fourth Stokes parameters over typical soil and canopy scenes are far smaller than T_{Bv} and T_{Bh}. Therefore, the LDCR antenna temperatures T_U and T_D can be approximately expressed as

$$\begin{aligned} T_{U/D} &= \frac{1}{4\pi}\iint_{2\pi}[G_{vv,U/D}(\theta,\phi)T_{B_v}(\theta,\phi) \\ &\quad + G_{hh,U/D}(\theta,\phi)T_{B_h}(\theta,\phi)]\sin\theta d\theta d\phi \\ &\quad + \eta_l \frac{1}{4\pi}\iint_{2\pi}\left[G_{vv,U/D}(\theta,\phi) + G_{hh,U/D}(\theta,\phi)\right] \\ &\quad \times T_{\downarrow}(\theta)\sin\theta d\theta d\phi \\ &\quad + (1-\eta_l)T_{P_A}. \end{aligned} \quad (21)$$

The direct solar contribution to the LDCR downwelling antenna temperature T_D is included as follows:

$$\begin{aligned} T_D &= \eta_l \frac{1}{4\pi}\iint_{\Omega_\odot}\left(G_{vv,D}(\theta,\phi) + G_{hh,D}(\theta,\phi)\right)(T_\odot - T_{\downarrow}) \\ &\quad \times \sin\theta d\theta d\phi \\ &\quad + \eta_l \frac{1}{4\pi}\iint_{2\pi}\left(G_{vv,D}(\theta,\phi) + G_{hh,D}(\theta,\phi)\right)T_{\downarrow}(\theta) \\ &\quad \times \sin\theta d\theta d\phi \\ &\quad + \eta_l \frac{1}{4\pi}\iint_{2\pi}\left[G_{vv,D}(\theta,\phi)T_{B_v}(\theta,\phi) + G_{hh,D}(\theta,\phi)\right. \\ &\quad \left.\times T_{B_h}(\theta,\phi)\right] \\ &\quad \sin\theta d\theta d\phi + (1-\eta_l)T_{P_A} \end{aligned} \quad (22)$$

where $T_\odot$ is the solar brightness temperature which ranges from 1×10^5 to 5×10^5 K at L-band [28], [29]. The solid angle $\Omega_\odot$ is calculated from the mean sun–earth distance of 1.496×10^8 km. The maximum direct solar contribution to the LDCR T_D thus ranges from 1.7 to 9 K and cannot be

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neglected. The direct solar and downwelling contribution can be operationally compensated based on solar zenith angle, solar cycle, cloud cover condition, sUAS pitch and roll angle, and MiCo antenna radiation patterns.

The galactic background contribution to LDCR T_D can be modeled using the galactic noise map at L-band provided by Le Vine *et al.* [30], [47], along with declination and right ascension. This contribution is calculated to be less than 0.5 K to the band-averaged LDCR received downwelling antenna temperature.

B. Dielectric Mixing Model

Using the soil dielectric mixing model of Dobson *et al.* [20], [21], the effective soil dielectric constant can be written as a function of soil moisture and soil texture

$$\epsilon_s = \epsilon_s' - j\epsilon_s'' \tag{23}$$

$$\epsilon_s' = \left[1 + \frac{\rho_b}{\rho_r}\epsilon_r^{\alpha} + m_v^{\beta'}\epsilon_{fw}'^{\alpha} - m_v\right]^{1/\alpha} \tag{24}$$

$$\epsilon_s'' = \left[m_v^{\beta''}\epsilon_{fw}''^{\alpha}\right]^{1/\alpha} \tag{25}$$

where the dielectric constant parameter ϵ_r is calculated as

$$\epsilon_r = (1.01 + 0.44\rho_r)^2 - 0.062 \tag{26}$$

and m_v is the VSM fraction, $\rho_b = 1.4$ g/cm^3 is the bulk soil density, and $\rho_r = 2.66$ g/cm^3 is the density of the solid soil particles. The parameter α is an empirical constant equal to 0.65, and β' and β'' are calculated as

$$\beta' = 1.2478 - 0.519S - 0.152C \tag{27}$$

$$\beta'' = 1.33797 - 0.603S - 0.166C \tag{28}$$

where S and C are the fractions of sand and clay (respectively) in the soil. The water dielectric constants ϵ_{fw}' and ϵ_{fw}'' are calculated using the Debye model [46]

$$\epsilon_{fw}' = \epsilon_{w\infty} + \frac{\epsilon_{w0} - \epsilon_{w\infty}}{1 + (2\pi f \tau_w)^2} \tag{29}$$

$$\epsilon_{fw}'' = \frac{2\pi f \tau_w(\epsilon_{w0} - \epsilon_{w\infty})}{1 + (2\pi f \tau_w)^2} + \frac{\sigma_{eff}}{2\pi f \epsilon_0}\frac{\rho_r - \rho_b}{\rho_r m_v} \tag{30}$$

where f is the frequency, ϵ_{w0} is the static dielectric constant and $\epsilon_{w\infty}$ is the dielectric constant of water in the high-frequency limit, τ_w is relaxation time of water, and σ_{eff} is the soil conductivity calculated as

$$\sigma_{\text{eff}} = -1.645 + 1.939\rho_b - 2.25622S + 1.594C \tag{31}$$

where ρ_b is in g/cm^3. The conductivity σ_{eff} depends on soil constitution and bulk density.

C. SVRT Surface Roughness Correction

For both SMAP and SMOS soil moisture retrieval algorithms, the surface roughness correction is performed based on the semiempirical H-Q-N method of Wang and Choudhury [31]–[35]. The original H-Q-N model uses two primary parameters: h_{R_p} and Q_R. An additional parameter N_{R_p} was introduced in subsequent investigations to better model multiangular and dual-polarization data [36]. Using H-Q-N, the rough soil reflectivities r_v^* and r_h^* for v and h polarizations are calculated as

$$r_v^*(\theta) = [(1 - Q_R(\theta))r_v(\theta) + Q_R(\theta)r_h(\theta)]e^{-h_{R_v}(\cos\theta)^{N_{R_v}}} \tag{32}$$

$$r_h^*(\theta) = [(1 - Q_R(\theta))r_h(\theta) + Q_R(\theta)r_v(\theta)]e^{-h_{R_h}(\cos\theta)^{N_{R_h}}} \tag{33}$$

where h_{R_p} is a semiempirical surface roughness parameter accounting for both coherent and incoherent scattering effects, Q_R is a polarization mixing parameter, and N_{R_p} is an angular exponent [14], [36].

The H-Q-N model was incorporated in the LDCR SVRT model under the assumptions of low vegetation cover for which $h_R = h_{R_v} = h_{R_h}$, $Q_R = 0$, and $N_{R_v} = N_{R_h} = 2$. These quantities follow the SMAP algorithm based on independent VWC data [14]. Given these assumptions, the rough surface soil reflectivities are

$$r_v^*(\theta) = r_v(\theta)e^{-h_R\cos^2(\theta)} \tag{34}$$

$$r_h^*(\theta) = r_h(\theta)e^{-h_R\cos^2(\theta)}. \tag{35}$$

For the SVRT vegetation correction and surface roughness corrections, the parameters ω_p, b_p, and h_R are tabulated as functions of land cover types where ω_p is assumed to be independent of polarization (i.e., $\omega_h = \omega_v$ [15], [27] (Table II). For vegetation covers of corn, wheat, soybean and tall grass, the ω_p, b_h, and b_v values are taken to be the best-fit values provided by Wigneron *et al.* [27] with an average of three published values used the albedo of corn. The ω_p, b_h, and b_v values for land cover types of short grass, bare soil, and forests are set to be the same as used in the current SMAP baseline retrieval algorithm [15]. For grain sorghum and sugar beets, which have been sparsely studied, the values of ω_p and b_p are set to be the same as for wheat and soybean (respectively). For built land cover, the scattering albedo ω_p ranges from 0.4 to 0.95 depending on building surface type (e.g., metal roof and concrete pad), and canopy attenuation is set to infinity to ensure that the soil moisture underneath is effectively immeasurable [37]. In this study, the metal-roofed IRF site buildings were assigned an albedo of 0.95. The surface roughness correction parameter h_R is set to be the same as used in the current SMAP baseline retrieval algorithm for all land cover types [15].

V. T_P Mapping Methods

In the full-domain soil moisture retrieval algorithm, the antenna temperature measurements at all sUAS sampling points are related to the soil moisture at all user-defined mapping grid points using an observation matrix. In order get this observation matrix, surface physical temperature values at the user-defined mapping grid points were optimally determined using an unbiased statistical LMMSE estimator. This estimator mapped from the irregularly sampled flight data grid to a regular mapping grid [38], [40], [41], [49]. This method is effectively identical to what is termed ordinary Kriging [39]. Letting $\overline{y}$ be the measured surface physical temperature T_P at the trajectory sample points and $\overline{x}$ be the true values of T_P at a user-defined rectangular grid of points, the relationship between these quantities can be expressed as

$$\overline{y} = \overline{\overline{W}}_f\overline{x} + \overline{n} \tag{36}$$

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where $\overline{n}$ is the instrument observation noise and $\overline{\overline{W}}_f$ is the flight sampling operator. This operator is effectively a sparse matrix wherein only a small number of the T_P values appear in any measured sample. Since the mean values of $\overline{y}$ and $\overline{x}$ are nonzero the estimated value of $\overline{x}$ will be

$$\widehat{\overline{x}} = \langle \overline{x} \rangle + \overline{\overline{D}}_f(\overline{y} - \langle \overline{y} \rangle). \tag{37}$$

Based on the orthogonality principle [12], the operator $\overline{\overline{D}}$ for statistical LMMSE is readily found to be

$$\overline{\overline{D}}_f = \overline{\overline{R}}_{xy}\overline{\overline{R}}_{yy}^{-1} \tag{38}$$

where $\overline{\overline{R}}_{xy}$ is the covariance matrix of T_P for all pairs of user grid and observed sample points while $\overline{\overline{R}}_{yy}$ is the covariance matrix of T_P values at the sample points. In order to ensure that the LMMSE is unbiased, (38) is modified as

$$\left[\overline{\overline{D}}_f, \overline{\lambda}_x\right] \times \begin{bmatrix} \overline{\overline{R}}_{yy} & \overline{1}_y \\ \overline{1}_y^t & 0 \end{bmatrix} = \left[\overline{\overline{R}}_{xy}, \quad \overline{1}_x\right] \tag{39}$$

where $\overline{\lambda}$ is the Lagrange multiplier vector, and $\overline{1}_\alpha$ is a vector of one of dimension of the associated parameters. There is, in general, one Lagrange multiplier per user grid point. It is assumed that in a local neighborhood of each grid point, the mean values of T_P at the sample points are the same as the mean values of T_P at corresponding user grid point. This assumption is reasonable because the surface physical temperature can be regarded as a gradually spatially varying function. For an unbiased LMMSE operator (37) thus becomes

$$\widehat{\overline{x}} = \overline{\overline{D}}_f \overline{y} \tag{40}$$

where the mean values are not explicitly needed, and the estimation operator $\overline{\overline{D}}$ is modified from (38) as

$$\overline{\overline{D}}_f = \overline{\overline{R}}_{xy}\overline{\overline{R}}_{yy}^{-1}\left(\overline{\overline{I}} - \frac{\overline{1}_y\overline{1}_y^t\overline{\overline{R}}_{yy}^{-1}}{\overline{1}_y^t\overline{\overline{R}}_{yy}^{-1}\overline{1}_y}\right) + \frac{\overline{1}_x\overline{1}_y^t\overline{\overline{R}}_{yy}^{-1}}{\overline{1}_y^t\overline{\overline{R}}_{yy}^{-1}\overline{1}_y}. \tag{41}$$

The surface physical temperature can be treated as geostatistical data with a spatially homogeneous radial covariance function that can be used to calculate $\overline{\overline{R}}_{xy}$ and $\overline{\overline{R}}_{yy}$ [39], [41], [42]. The T_P correlation length depends on land cover type and cloud cover conditions. Land surface temperature covariance at high spatial resolution (i.e., decameter to meters) has been sparsely studied. In this study, the effective correlation length is chosen to be 85 m, so as to be consistent with the VSM correlation length used in the LDCR VSM retrieval algorithm [42]. Selecting such a limited correlation length ensures that the physical temperature sample and the physical temperature at the estimation point have the same (or nearly so) mean values.

The T_P values at user-defined mapping grid points have an estimation error covariance using this unbiased LMMSE estimation method of

$$\overline{\overline{R}}_{ee} = \overline{\overline{R}}_{xx} - \overline{\overline{R}}_{xy}\overline{\overline{R}}_{yy}^{-1}\overline{\overline{R}}_{yx} + \overline{\lambda}\overline{1}_y^t\overline{\overline{R}}_{yy}^{-1}\overline{1}_y\overline{\lambda}^t. \tag{42}$$

It can be proven that the unbiased LMMSE error is no lower than that of biased LMMSE error $\overline{\overline{R}}_{xx} - \overline{\overline{R}}_{xy}\overline{\overline{R}}_{yy}^{-1}\overline{\overline{R}}_{yx}$. The estimated T_P and its estimation error are inputs into the LDCR soil moisture retrieval algorithm.

VI. LDCR VSM Retrieval Algorithm

The retrieval of VSM on a user-defined mapping grid from LDCR $T_A \triangleq (T_U - (1-\eta_l)T_{P_A})/\eta_l$ samples uses a physical LMMSE estimator, which has been discussed in the context of remote sensing since the 1970s [22]. Let the T_A measurements at the sample locations on the full observation domain be an observed vector $\overline{y}$ and the true value of VSM on a user-defined (e.g. rectilinear) grid be an unknown vector $\overline{x}$. The relationship between $\overline{y}$ and $\overline{x}$ can be linearly approximated as

$$\overline{y} - \langle \overline{y} \rangle = \overline{W}\left(\overline{x}, \overline{P}\right) + \overline{n} \cong (\overline{\overline{W}}_o + \delta\overline{\overline{W}})(\overline{x} - \langle \overline{x} \rangle) + \overline{n} \tag{43}$$

where $\overline{W}(\overline{x}, \overline{P})$ is the forward observation operator, $\overline{P}$ is a possibly large parameter vector (to be described) affecting the determination of $\overline{W}$, $\overline{\overline{W}}_o = \overline{\overline{W}}\big|_{\langle \overline{x}, \overline{P} \rangle}$ is the value of the Jacobian $(\partial\overline{W}/\partial\overline{x})$ at the expected value of $\overline{x}$ and $\overline{P}$, and $\overline{n}$ is the observation noise.

The operator $\overline{W}$ is a weakly nonlinear function of the expected vegetation opacity and surface roughness, soil type and texture parameters, and also depends on the antenna radiation pattern and the position vector from each sampling point to each mapping point. A random component $\delta\overline{\overline{W}}$ results from all parameter uncertainties, and includes small nonlinear terms in the radiative transfer model

$$\delta\overline{\overline{W}} = \sum_i \frac{\partial\overline{\overline{W}}}{\partial\overline{P}_i}\bigg|\langle \overline{P}_i \rangle \delta\overline{P}_i + \frac{1}{2}\frac{\partial\overline{\overline{W}}}{\partial\overline{x}}\bigg|\langle \overline{x} \rangle(\overline{x} - \langle \overline{x} \rangle) \tag{44}$$

where the parameters in $\overline{P}$ fall into two categories: geophysical measurements and empirical parameters. Geophysical measurements include VWC W_v, surface physical temperature T_P, and soil sand and clay percentages S and C. Empirical parameters include b_h, b_v, ω_p, and surface roughness parameter h_R.

Since the mean values of $\overline{y}$ and $\overline{x}$ are nonzero, the estimated value of $\overline{x}$ will be

$$\widehat{\overline{x}} = \overline{\overline{D}}_s(\overline{y} - \langle \overline{y} \rangle) + \langle \overline{x} \rangle \tag{45}$$

where the physical LMMSE operator $\overline{\overline{D}}_s$ is readily found to be [12]

$$\overline{\overline{D}}_s = \overline{\overline{R}}_{xx}\overline{\overline{W}}_o^t (\overline{\overline{W}}_o\overline{\overline{R}}_{xx}\overline{\overline{W}}_o^t + \langle \delta\overline{\overline{W}}(\overline{x} - \langle \overline{x} \rangle)(\overline{x} - \langle \overline{x} \rangle)^t\delta\overline{\overline{W}}^t \rangle + \overline{\overline{R}}_{nn})^{-1}. \tag{46}$$

In the above the $\overline{\overline{R}}_{xx}$ is the spatial covariance matrix of soil moisture on the user grid and $\overline{\overline{R}}_{nn}$ is the observation error covariance matrix that is only dependent on the LDCR radiometric noise. Assuming the antenna temperature errors at different sampling points are uncorrelated, it follows that $\overline{\overline{R}}_{nn} = \Delta T_{\text{rms},s}^2\overline{\overline{I}}$, where $\Delta T_{\text{rms},s}^2$ is discussed in (12).

A modified observation error covariance matrix $\overline{\overline{R}}'_{nn}$ incorporating all uncertain quantities in (46) can now be defined as $\overline{\overline{R}}'_{nn} = \langle \delta\overline{\overline{W}}\ (\overline{x} - \langle \overline{x} \rangle)(\overline{x} - \langle \overline{x} \rangle)^t\delta\overline{\overline{W}}^t \rangle + \overline{\overline{R}}_{nn}$. This modified covariance accounts for vegetation and surface roughness correction parameter uncertainties, temperature estimation error at the mapping grid point, soil texture information uncertainties,

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observation operator nonlinearities, and radiometric measurement noise, as follows:

$$\begin{aligned}
\overline{\overline{R}}'_{nn} &= \left\langle \frac{\partial \overline{\overline{W}}}{\partial \overline{W}_v}\Big|_{\langle \overline{W}_v\rangle} \delta \overline{W}_v (\overline{x}-\langle \overline{x}\rangle)(\overline{x}-\langle \overline{x}\rangle)^t \left(\frac{\partial \overline{\overline{W}}}{\partial \overline{W}_v}\Big|_{\langle \overline{W}_v\rangle} \delta \overline{W}_v\right)^t \right\rangle \\
&+ \left\langle \frac{\partial \overline{\overline{W}}}{\partial \overline{T}_P}\Big|_{\langle \overline{T}_P\rangle} \delta \overline{T}_P (\overline{x}-\langle \overline{x}\rangle)(\overline{x}-\langle \overline{x}\rangle)^t \left(\frac{\partial \overline{\overline{W}}}{\partial \overline{T}_P}\Big|_{\langle \overline{T}_P\rangle} \delta \overline{T}_P\right)^t \right\rangle \\
&+ \left\langle \frac{\partial \overline{\overline{W}}}{\partial \overline{S}}\Big|_{\langle \overline{S}\rangle} \delta \overline{S} (\overline{x}-\langle \overline{x}\rangle)(\overline{x}-\langle \overline{x}\rangle)^t \left(\frac{\partial \overline{\overline{W}}}{\partial \overline{S}}\Big|_{\langle \overline{S}\rangle} \delta \overline{S}\right)^t \right\rangle \\
&+ \left\langle \frac{\partial \overline{\overline{W}}}{\partial \overline{C}}\Big|_{\langle \overline{C}\rangle} \delta \overline{C} (\overline{x}-\langle \overline{x}\rangle)(\overline{x}-\langle \overline{x}\rangle)^t \left(\frac{\partial \overline{\overline{W}}}{\partial \overline{C}}\Big|_{\langle \overline{C}\rangle} \delta \overline{C}\right)^t \right\rangle \\
&+ \left\langle \frac{\partial \overline{\overline{W}}}{\partial \overline{b}_h}\Big|_{\langle \overline{b}_h\rangle} \delta \overline{b}_h (\overline{x}-\langle \overline{x}\rangle)(\overline{x}-\langle \overline{x}\rangle)^t \left(\frac{\partial \overline{\overline{W}}}{\partial \overline{b}_h}\Big|_{\langle \overline{b}_h\rangle} \delta \overline{b}_h\right)^t \right\rangle \\
&+ \left\langle \frac{\partial \overline{\overline{W}}}{\partial \overline{b}_v}\Big|_{\langle \overline{b}_v\rangle} \delta \overline{b}_v (\overline{x}-\langle \overline{x}\rangle)(\overline{x}-\langle \overline{x}\rangle)^t \left(\frac{\partial \overline{\overline{W}}}{\partial \overline{b}_v}\Big|_{\langle \overline{b}_v\rangle} \delta \overline{b}_v\right)^t \right\rangle \\
&+ \left\langle \frac{\partial \overline{\overline{W}}}{\partial \overline{\omega}_p}\Big|_{\langle \overline{\omega}_p\rangle} \delta \overline{\omega}_p (\overline{x}-\langle \overline{x}\rangle)(\overline{x}-\langle \overline{x}\rangle)^t \left(\frac{\partial \overline{\overline{W}}}{\partial \overline{\omega}_p}\Big|_{\langle \overline{\omega}_p\rangle} \delta \overline{\omega}_p\right)^t \right\rangle \\
&+ \left\langle \frac{\partial \overline{\overline{W}}}{\partial \overline{h}_R}\Big|_{\langle \overline{h}_R\rangle} \delta \overline{h}_R (\overline{x}-\langle \overline{x}\rangle)(\overline{x}-\langle \overline{x}\rangle)^t \left(\frac{\partial \overline{\overline{W}}}{\partial \overline{h}_R}\Big|_{\langle \overline{h}_R\rangle} \delta \overline{h}_R\right)^t \right\rangle \\
&+ \frac{1}{4}\left\langle \frac{\partial \overline{\overline{W}}}{\partial \overline{x}}\Big|_{\langle \overline{x}\rangle} (\overline{x}-\langle \overline{x}\rangle)(\overline{x}-\langle \overline{x}\rangle)(\overline{x}-\langle \overline{x}\rangle)^t \right. \\
&\qquad \left. \times \left(\frac{\partial \overline{\overline{W}}}{\partial \overline{x}}\Big|_{\langle \overline{x}\rangle} (\overline{x}-\langle \overline{x}\rangle)\right)^t \right\rangle \\
&+ \Delta T^2_{\mathrm{rms},s}\, \overline{\overline{I}} \qquad (47)
\end{aligned}$$

with the assumption that all parameter deviations are uncorrelated.

A. Spatial VSM Covariance

The covariance of soil moisture at any two mapping grid points is in general nonzero. This is particularly true for cropland areas managed using uniform field irrigation practices. One way to calculate the VSM covariance matrix is to use a stationary spatial covariance function [38]–[40]

$$R_{xx}(\overline{r}_1,\overline{r}_2) = \begin{cases} \sigma_x^2\left(1-\dfrac{3}{2}\left(\dfrac{|\overline{r}_1-\overline{r}_2|}{a}\right)+\dfrac{1}{2}\left(\dfrac{|\overline{r}_1-\overline{r}_2|}{a}\right)^3\right) \\ \qquad |\overline{r}_1-\overline{r}_2| \le a \\ 0 \\ \qquad |\overline{r}_1-\overline{r}_2| > a \end{cases} \qquad (48)$$

where σ_x^2 is the VSM variance. In the following soil moisture retrieval performance calculations, a value of σ_{x_i} of 8.3% is used at all mapping grid points. This value of σ_{x_i} presumes a Gaussian VSM distribution with 6-sigma range from 0% to 50% VSM. Equation (48) is effectively a VSM semi-variogram dependent only on the separation distance between two points. The value of a depends on crop land management practices, soil texture, and topography, and possibly the VSM itself [42]. In this study, a is set to be approximately 85 m based on a least square fit to an empirical correlation function provided by Western *et al.* [42].

For crop land, a modified land section covariance (LSC) function based on managed land sections (e.g., irrigated or tilled) can further be specified. In this case, covariance is calculated using (49) when $\overline{r}_1$ and $\overline{r}_2$ are within the same crop section, that is for land with the same vegetation cover type, irrigation, tillage, and fertilization practices, and general cultivation state. When $\overline{r}_1$ and $\overline{r}_2$ are in different sections, the covariance is set to zero.

A third method for generating the VSM covariance matrix is to assume that the VSM correlation is unity within the same crop section and thus $R_{xx}(\overline{r}_1,\overline{r}_2)=\sigma_x^2$. When $\overline{r}_1$ and $\overline{r}_2$ are in different sections the covariance is set to zero. The above is effectively a uniform correlation LSC (denoted as UCLSC) model. By enforcing unity correlation, the resulting VSM estimate is constant within any given crop section. While this condition detracts from identifying subsection spatial variations in VSM, it also results in a more accurate section-averaged value of VSM. The improved accuracy results from the weighting of all available LDCR observations in determining a single section-averaged VSM value.

B. Spatial VSM-T_A Operator $\overline{\overline{W}}_o$

In order to determine the linear observation operator $\overline{\overline{W}}_o$ which relates the T_A at any sampling point to VSM at any user-defined mapping point, the LDCR measured T_A at the k^{th} sampling point T_{A_k} can be rewritten from the integral equation (20) to second order in VSM as

$$\begin{aligned}
T_{A_k} &= T_{A_{k,s}} + \sum_{i=1}^{M} T_{A_{k,i}} \cong T_{A_{k,s}} \\
&\quad + \sum_{i=1}^{M}\Big(W_{o_{k,i}}(m_{v_i}-\langle m_{v_i}\rangle) + W'_{o_{k,i}}(m_{v_i}-\langle m_{v_i}\rangle)^2 \\
&\qquad + \langle T_{A_{k,i}}\rangle\Big) \qquad (49)
\end{aligned}$$

where $T_{A_{k,i}}$ is the antenna temperature contribution, $\langle T_{A_{k,i}}\rangle$ is its mean value at the ith user-defined mapping point, m_{v_i} and $\langle m_{v_i}\rangle$ are the VSM and its mean value (respectively) at this mapping point, M is the total number of user mapping grid points, and $T_{A_{k,s}}$ is the combined sky and solar downwelling antenna temperature contribution. The values of $T_{A_{k,i}}$ and $T_{A_{k,s}}$ can be calculated from (20) separately as

$$\begin{aligned}
T_{A_{k,i}} &= \frac{1}{4\pi}\iint_{\delta\theta_{k,i},\delta\phi_{k,i}} \big(G_{vv,U}(\theta,\phi)T_{B_v}(\theta,\phi) \\
&\qquad + G_{hh,U}(\theta,\phi)T_{B_h}(\theta,\phi)\big)\sin\theta d\theta d\phi \qquad (50)
\end{aligned}$$

$$\begin{aligned}
T_{A_{k,s}} &= \frac{1}{4\pi}\iint_{2\pi}\big(G_{vv,U}(\theta,\phi)+G_{hh,U}(\theta,\phi)\big)T_{\downarrow}(\theta,\phi) \\
&\quad \times \sin\theta d\theta d\phi \\
&\quad + \frac{1}{4\pi}\iint_{\Omega_\odot}\big(G_{vv,U}(\theta,\phi)+G_{hh,U}(\theta,\phi)\big)(T_\odot - T_{\downarrow}) \\
&\quad \times \sin\theta d\theta d\phi \qquad (51)
\end{aligned}$$

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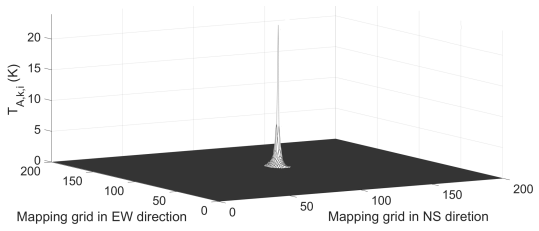

Fig. 7. Antenna temperature contribution from all mapping grid points for the (e.g.) 800th sampling point given a VSM of 25%.

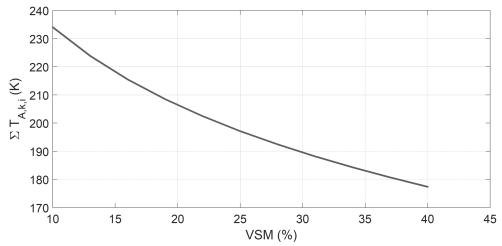

Fig. 8. Antenna temperature contribution sum of all mapping grid points as a functions of VSM.

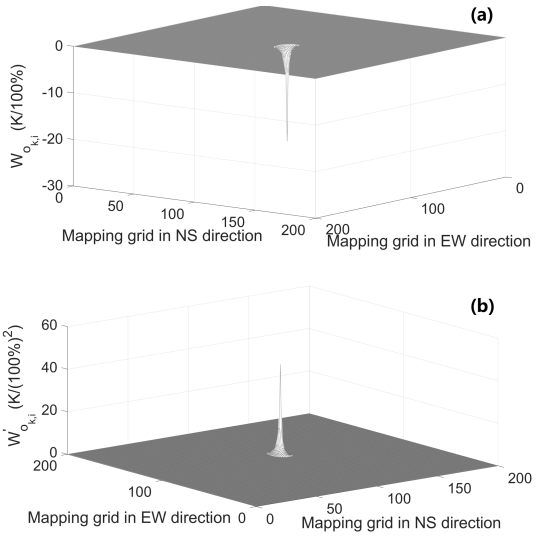

Fig. 9. Second-order polynomial fit parameters (a) $W_{o_{k,i}}$ and (b) $W'_{o_{k,i}}$ for the antenna temperature contributions at the $k = 800^{\text{th}}$ sampling point.

where $\delta\theta_{k,i}$ and $\delta\phi_{k,i}$ are determined by the vector between the ith user mapping point and the kth sampling point along with sUAS flight direction, altitude, and pitch and roll angle at the kth sampling point.

Taking the 800th LDCR sampling point as an example, and assuming T_P is 298 K and corn vegetation cover, the $T_{A_{800,i}}$ values are calculated and plotted assuming a VSM of 25% (Fig. 7). The figure is essentially the spatial sensitivity pattern of the sUAS sensor. It is noted that more than 95% percent of the antenna temperature contribution is from mapping points located within a circle of ∼50 m diameter. This range is consistent with LDCR MiCo antenna foot print size given the sUAS flight altitude of ∼50 m and the LDCR MiCo antenna 3-dB beamwidth of ∼45°. The sum of the antenna temperature contribution from all mapping points $\sum_{i=1}^{M} T_{A_{k,i}}$ is plotted for various VSM values in Fig. 8. The sum decreases from 235 to 178 K as the VSM increases from 10% to 40%. This calculation indicates a dynamic range of 95 K over the full 0%–50% VSM range.

In (50), the values of $W_{o_{k,i}}$ and $W'_{o_{k,i}}$ were determined using a second order polynomial fit and are plotted for the $k = 800^{\text{th}}$ sampling point (Fig. 9). It is noted that the $W_{o_{k,i}}$ values are negative with largest minimum value of ∼-20 K/100%, and the parameter $W'_{o_{k,i}}$ has a maximum value of ∼42 K/(100%)^2. The $W_{k,i}$ nonlinearity is accounted for as a model error in (43) as $W'_{o_{k,i}} = (1/2)(\partial W_{k,i}/\partial m_{v_i})\big|_{\langle m_{v_i}\rangle}$, and $(\partial W_{k,i}/\partial m_{v_j}) = 0$ when $i \neq j$. The second-order component in (50) is thus found to be less than 2% of the antenna temperature contribution for most of the mapping grid.

The partial derivatives of $\overline{\overline{W}}$ with respect to $\overline{P}$ are determined similarly. Variations in any parameter are used to calculate the value of $W_{k,i}$, and partial derivatives $(\partial W_{k,i}/\partial P_i)$ are then determined using a linear fit of $W_{k,i}$ to variations in any parameter P_i (respectively) while the other of these parameter values remain fixed. Similarly, values of $(\partial W_{k,i}/\partial \overline{P}_j)$ are set to zero so that the estimation of $\overline{P_i}$ at the i^{th} mapping point only effects the determination of $\overline{\overline{W}}$ relative to i^{th} mapping point. Once $\overline{\overline{W}}_o$ and $\delta\overline{\overline{W}}$ are determined, the LDCR VSM at all mapping grid points can be estimated using (45)–(46).

C. VSM Estimation Error

The VSM estimation error covariance for the full mapping grid using the physical LMMSE is calculated as

$$\overline{\overline{R}}_{ee} = \left(\overline{\overline{I}} - \overline{\overline{D}}_s\ \overline{\overline{W}}_o\right)\overline{\overline{R}}_{xx}\left(\overline{\overline{I}} - \overline{\overline{D}}_s\ \overline{\overline{W}}_o\right)^t + \overline{\overline{D}}_s\ \overline{\overline{R}}'_{nn}\overline{\overline{D}}^t_s. \quad (52)$$

As shown in (48), this estimation error depends on $\Delta T_{\text{rms},s}$ along with the estimation errors of all model parameters. It is assumed that for any parameter P at any mapping point, the estimation error includes estimation bias and variation as

$$\delta P = \beta_{e_P} + \delta_{e_P}. \quad (53)$$

Assuming that the bias correlates within a crop land section, β_{e_P} has a uniform value within each such section. The parameter error variations δ_{e_P} at different mapping grid points are assumed to be uncorrelated, with standard deviation σ_{e_P}. For the empirical parameters in Table II it is reasonable to assume σ_{e_P} to be zero, but not β_{e_P}. In addition to these assumptions, assuming VSM process variations are uncorrelated the observation error covariance matrix consequently becomes conservatively bounded by

$$\begin{aligned}\overline{\overline{R}}'_{nn} &= \sum_{i=1}^{M}\sigma^2_{e_{W_{v_i}}}\sigma^2_x\left\langle\frac{\partial\overline{\overline{W}}}{\partial W_{v_i}}\left(\frac{\partial\overline{\overline{W}}}{\partial W_{v_i}}\right)^t\right\rangle \\ &+ \sum_{i=1}^{M}\sigma^2_{e_{T_{P_i}}}\sigma^2_x\left\langle\frac{\partial\overline{\overline{W}}}{\partial T_{P_i}}\left(\frac{\partial\overline{\overline{W}}}{\partial T_{P_i}}\right)^t\right\rangle \\ &+ \sum_{i=1}^{M}\sigma^2_{e_{S_i}}\sigma^2_x\left\langle\frac{\partial\overline{\overline{W}}}{\partial S_i}\left(\frac{\partial\overline{\overline{W}}}{\partial S_i}\right)^t\right\rangle + \sum_{i=1}^{M}\sigma^2_{e_{C_i}}\sigma^2_x\left\langle\frac{\partial\overline{\overline{W}}}{\partial C_i}\left(\frac{\partial\overline{\overline{W}}}{\partial C_i}\right)^t\right\rangle\end{aligned}$$

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$$+\sum_{l=1}^{L}\beta_{e_{W_{v_l}}}^2\sigma_x^2\left\langle\frac{\partial\overline{\overline{W}}}{\partial\overline{W}_{v_l}}\overline{1}_{M_l}\overline{1}_{M_l}^t\left(\frac{\partial\overline{\overline{W}}}{\partial\overline{W}_{v_l}}\right)^t\right\rangle$$
$$+\sum_{l=1}^{L}\beta_{e_{T_{P_l}}}^2\sigma_x^2\left\langle\frac{\partial\overline{\overline{W}}}{\partial\overline{T}_{P_l}}\overline{1}_{M_l}\overline{1}_{M_l}^t\left(\frac{\partial\overline{\overline{W}}}{\partial\overline{T}_{P_l}}\right)^t\right\rangle$$
$$+\sum_{l=1}^{L}\beta_{e_{S_l}}^2\sigma_x^2\left\langle\frac{\partial\overline{\overline{W}}}{\partial\overline{S}_l}\overline{1}_{M_l}\overline{1}_{M_l}^t\left(\frac{\partial\overline{\overline{W}}}{\partial\overline{S}_l}\right)^t\right\rangle$$
$$+\sum_{l=1}^{L}\beta_{e_{C_l}}^2\sigma_x^2\left\langle\frac{\partial\overline{\overline{W}}}{\partial\overline{C}_l}\overline{1}_{M_l}\overline{1}_{M_l}^t\left(\frac{\partial\overline{\overline{W}}}{\partial\overline{C}_l}\right)^t\right\rangle$$
$$+\sum_{l=1}^{L}\beta_{e_{b_{h_l}}}^2\sigma_x^2\left\langle\frac{\partial\overline{\overline{W}}}{\partial\overline{b}_{h_l}}\overline{1}_{M_l}\overline{1}_{M_l}^t\left(\frac{\partial\overline{\overline{W}}}{\partial\overline{b}_{h_l}}\right)^t\right\rangle$$
$$+\sum_{l=1}^{L}\beta_{e_{b_{v_l}}}^2\sigma_x^2\left\langle\frac{\partial\overline{\overline{W}}}{\partial\overline{b}_{v_l}}\overline{1}_{M_l}\overline{1}_{M_l}^t\left(\frac{\partial\overline{\overline{W}}}{\partial\overline{b}_{v_l}}\right)^t\right\rangle$$
$$+\sum_{l=1}^{L}\beta_{e_{\omega_{p_l}}}^2\sigma_x^2\left\langle\frac{\partial\overline{\overline{W}}}{\partial\overline{\omega}_{p_l}}\overline{1}_{M_l}\overline{1}_{M_l}^t\left(\frac{\partial\overline{\overline{W}}}{\partial\overline{\omega}_{p_l}}\right)^t\right\rangle$$
$$+\sum_{l=1}^{L}\beta_{e_{h_{R_l}}}^2\sigma_x^2\left\langle\frac{\partial\overline{\overline{W}}}{\partial\overline{h}_{R_l}}\overline{1}_{M_l}\overline{1}_{M_l}^t\left(\frac{\partial\overline{\overline{W}}}{\partial\overline{h}_{R_l}}\right)^t\right\rangle$$
$$+\frac{1}{4}\sum_{i=1}^{M}3\sigma_{x_i}^4\left\langle\frac{\partial\overline{\overline{W}}}{\partial x_i}\left(\frac{\partial\overline{\overline{W}}}{\partial x_i}\right)^t\right\rangle+\Delta T_{\mathrm{rms},s}^2\overline{\overline{I}} \tag{54}$$

where i is the mapping point number, l is the land section number, L is the total number of land sections, M_l is the total number of mapping points in the lth land section, and the following variances and biases are defined: $\sigma_{e_{W_{v_i}}}^2$ is the VWC error variance at the ith mapping grid point, $\beta_{e_{W_{v_l}}}$ is the VWC bias in the lth land section, $\frac{\partial\overline{\overline{W}}}{\partial\overline{W}_{v_l}}$ is the partial derivative of $\overline{\overline{W}}$ with respect to VWC W_{v_l} at all mapping grid points in the lth land section, and $\frac{\partial\overline{\overline{W}}}{\partial\overline{W}_{v_l}}\overline{1}_{M_l}=\frac{\partial\overline{W}}{\partial W_{v_1}}+\frac{\partial\overline{W}}{\partial W_{v_2}}+\ldots+\frac{\partial\overline{W}}{\partial W_{v_{M_l}}}$ is a matrix of size of $N\times M$ where N is the total number of sampling points. Spatial indexing using i and l is necessary since the values of parameter error variances at different mapping points or error biases in different land sections are not necessarily the same. It is noted that the uncorrelated estimation error covariances have smaller effects on VSM estimation error compared with the bias error. This smaller impact is a result of the application of the full-domain retrieval algorithm in that the observation operator $\overline{\overline{W}}_o$ relates T_A measurements at each sampling grid point to VSM at ~150 mapping grid points, and the VSM at each single mapping grid point is estimated from sUAS measurements at ~14 sampling points using the estimator $\overline{\overline{D}}_s$. Thus, the estimation error variance at each single mapping grid point is significantly reduced by the full-domain retrieval method. However, errors due to parameter bias are not reduced by this averaging process. This full-domain retrieval algorithm and estimation error are validated using experimental data in §VII and §VIII.

VII. Field Experiments in Canton OK

Two successful field experiments were performed at the Canton Soilscape site in September 2015 with LDCR Rev A integrated into the Tempest sUAS platform.

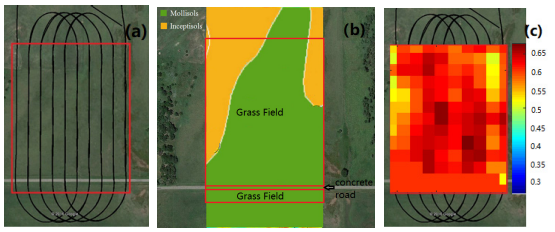

Fig. 10. (a) Flight trajectories, (b) soil type and land cover, and (c) VWC for the Canton OK site on September 8–9, 2015.

A. Sorties

Relevant images of the Canton site and flight trajectory overlays for the September 8th and 9th sorties illustrate the ~32 acre rectangular mapping area (Fig. 10). On these dates, Tempest was flown in serpentine trajectories with North-South (NS) flight lines. During the flight on September 8 with an 11:30 A.M. takeoff, cloud cover was ~60% and the ambient temperature was ~300 K. The test site was covered by dry green grass of ~7 in height. On September 9 with an 8:00 A.M. take off, the cloud cover was ~100% and the ambient temperature was ~293 K. The grass was lightly moistened with dew. No precipitation was observed on either test day. Each flight was ~16 min in duration. The VWC maps were generated using Landsat 7 Enhanced Thematic Mapper Plus (ETM+) data observed on September 5 with a spatial resolution of 30 m [43], [44].

B. LDCR Surface T_P Maps

The surface physical temperature was measured by the LDCR downlooking infrared (IR) sensor then mapped using the unbiased statistical LMMSE method with its estimation error. Taking the September 8th flight as an example, the measured surface T_P had a mean value of 302.9 K and standard deviation of 0.79 K during the mapping time period (~7 min). A minimum error of ~0.13 K beneath the flight lines and a maximum of ~0.32 K between flight lines are shown in Fig. 11. For the September 9th data, the measured T_P had a mean value of 292.9 K and a standard deviation of 0.16 K, and the T_P estimation error has a similar spatial variation although the values are lower by ~70% due to the lower overall T_P standard deviation. Due to the short duration of each flight there was likely a relatively small change in surface temperature between takeoff and landing.

C. LDCR Volumetric Soil Moisture

The LDCR VSM at the Canton site was retrieved using the full-domain algorithm from T_A measurement on the flight sampling grid along with estimated T_P maps [Fig. 12(a) and (b)]. A uniform spatial correlation function is used in this retrieval process to describe the vegetation cover homogeneity in the mapping area and the user-defined mapping grid size is ~5 × 5 m. Assuming that all parameters $\overline{P}$ are accurately known, the VSM estimation error standard deviations [shown in Fig. 12(c) and (d)] are calculated and found to be mainly dependent on the T_A estimation error and sUAS measurement locations. Minimum error is observed directly beneath

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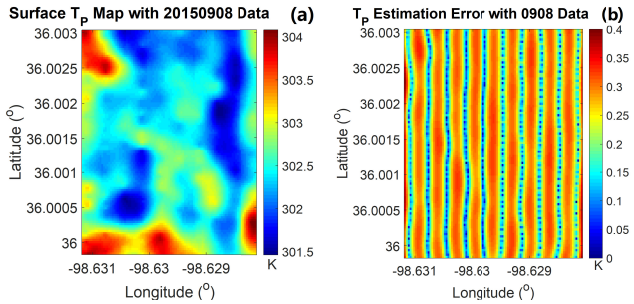

Fig. 11. Maps of (a) surface physical temperature T_P and (b) T_P estimation error at the Canton site on September 8, 2015.

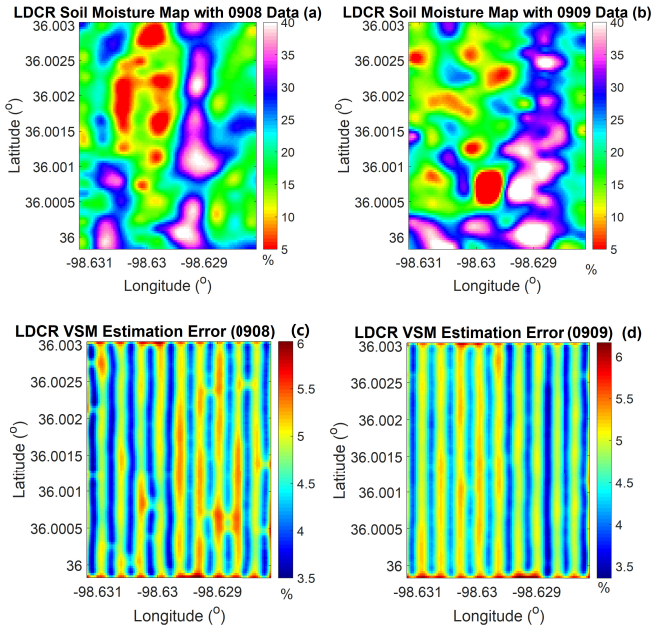

Fig. 12. LDCR retrieved VSM and VSM estimation error maps at the Canton site on (a) and (c) September 8 and (b) and (d) September 9.

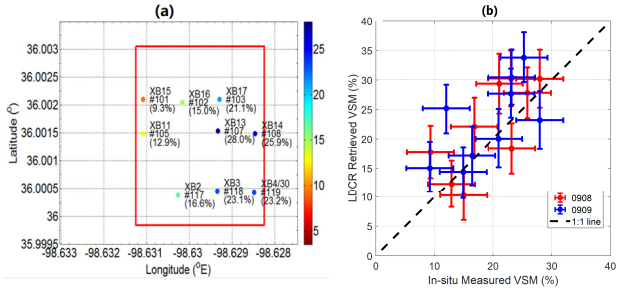

Fig. 13. LDCR Canton Soilscape *in situ* VSM measurements (a) averaged (September 8–9) values including probe locations, and (b) comparison with retrieved co-located LDCR VSM measurements.

flight lines to be ∼3.5%–4% VSM, and maximum error is ∼5%–5.5% VSM at locations between flight lines.

In situ measurements of VSM at the Canton site were available at nine locations in the LDCR mapping area [Fig. 13(a)] [6]. At each location, sensors were installed at three different depths below the soil surface (5, 15, and 30 cm), and sensor VSM measurement error was estimated to be 4%.

TABLE III
SOIL CLASSIFICATION AT IRF SITE

Soil Type	Description	Sand Percentage	Clay Percentage
3	Ascalon sandy loam, sandy substratu	70	15
5	Ascalon fine sandy loam	70	15
19	Haxtun sandy loam	75	10
23	Julesburg loamy sand	75	10
37	Rago loam	25	20

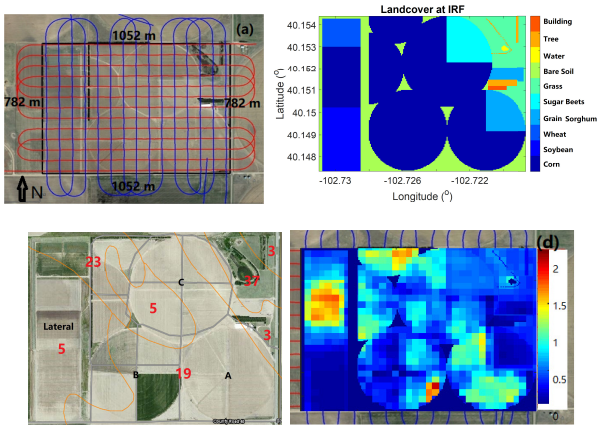

Fig. 14. Maps of (a) flight trajectories for June 14, 2016 (typical), (b) land cover type, (c) soil type, and (d) VWC at the IRF site.

The LDCR retrieved VSM values are compared with these measurements in Fig. 13(b) where a correlation coefficient of 0.70 is obtained.

VIII. FIELD EXPERIMENTS AT THE IRF SITE

Four successful field experiments were performed at the IRF site in Yuma, Colorado, in 2016 with LDCR Rev A and the Tempest sUAS.

A. Sorties

Two flights with lines in the NS direction (12:19 P.M. to 12:34 P.M.) and east-west (EW) direction (10:46 A.M. to 11:06 A.M.) on June 14, one flight with lines in the EW direction (12:50 P.M. to 13:07 P.M.) on June 20, and one flight with lines in the NS direction (10:04 A.M. to 10:22 A.M.) on June 23 were performed. The mapping area contained ten distinct vegetation covers including corn, soybean, wheat, grain sorghum, sugar beets, grass, standing water, trees, bare soil, and buildings (Fig. 14). The VWC maps were derived from Landsat 7 ETM+ data. Soil types in the mapping area were classified according to sand and clay percentage [Fig. 14(c), Table III].

B. LDCR Surface T_P Maps

Interpolated T_P and T_P estimation error maps for the two successive flights on June 14 are intercompared (Fig. 15). The two T_P maps are consistent, and the measured T_P for the noon

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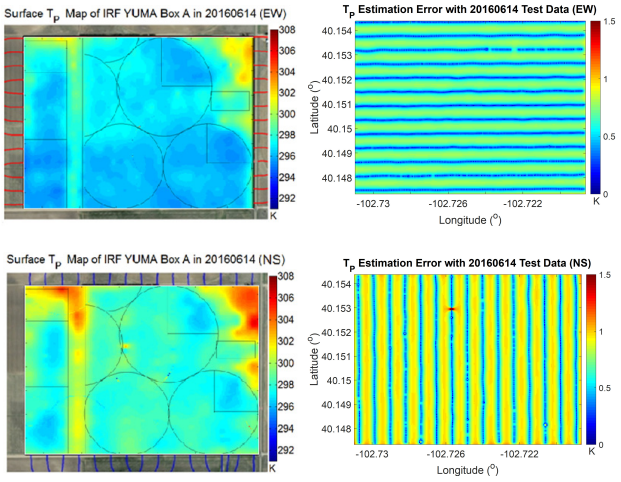

Fig. 15. (Left) Estimated T_P and (Right) T_P estimation error maps using the EW and NS flight data on June 14 at IRF site.

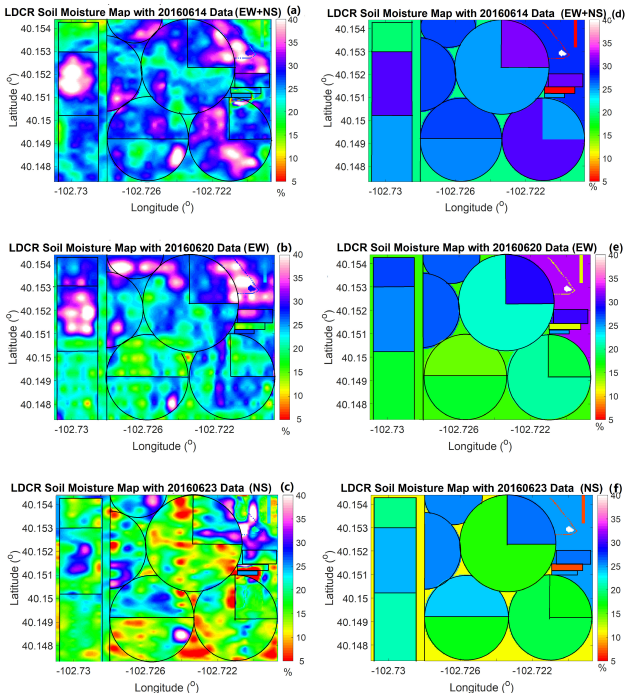

Fig. 16. LDCR VSM maps on (a) and (d) June 14, (b) and (e) June 20, and (c) and (f) June 23 using the full-domain LMMSE retrieval algorithm with a (Left) LSC function and (Right) UCLSC function.

(NS flight) is ∼3 K higher than that in the morning (EW flight). The surface T_P for bare soil and grass areas is higher than that for cropland by ∼5 K on both maps. The two T_P estimation error maps show similar spatial variation associated with the location of flight lines. For flights on June 20 and 23, T_P show higher values at bare soil and its estimation error show similar spatial variation. In all cases, the T_P error does not dominate the T_A error.

C. LDCR Volumetric Soil Moisture

LDCR VSM maps at the IRF site were retrieved using the full-domain LMMSE estimation method and the user-defined

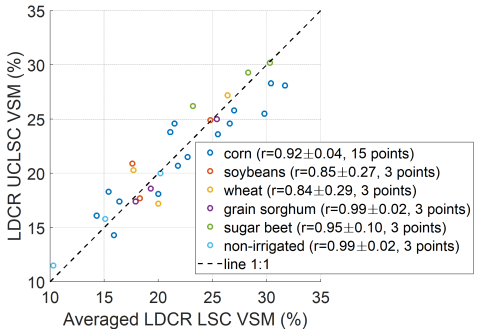

Fig. 17. Comparison between section-averaged LDCR LSC VSM and LDCR UCLSC VSM.

mapping grid size is ∼5 × 5 m. For the vegetation cover complexity and land management practices in this mapping area, both a land section covariance (LSC) function and uniform correlation LSC (UCLSC) function are used in this retrieval process, with results shown in Fig. 16. For the June 14th flights, the LDCR VSM are retrieved by combining both EW and NS flight data, and for June 20th and 23rd data, the VSM are retrieved from single flight direction data. As expected, the retrieved VSM using the LSC function show more detail in each land usage section while the VSM retrieved using the UCLSC function has a uniform value within each section. The VSM retrieved using the UCLSC function and average LSC VSM are highly correlated with a coefficient of 0.93 ± 0.03 for all vegetation cover types [48] (Fig. 17). Correlations for individual crop type are similarly high, although the number of samples available is small.

The UCLSC function is based on the assumption that the VSM within each land section has a unity correlation coefficient. The rationality of this assumption primarily depends on the land section size, soil texture homogeneity, and soil topography within the land section. UCLSC is anticipated to be useful in precision irrigation due to its weighted averaging of all available measurements over a section.

In situ measured soil moisture data was supplied by the IRF, however the VSM mapping error is difficult to quantify using these data. There are eight probes in the mapping area: probes 8–9 within the corn sector of circle B, probes 2–4 within the soybeans portion of the lateral area, and probes 5–7 within the sugar beet sector of circle C [Fig. 18(a)]. *In situ* measurements at different depths (10, 20, 30 cm) were recorded every 15 min. The *in situ* VSM at 10 cm is found to be consistent with IRF irrigation records and precipitation data history during which no precipitation was recorded from June 14 to June 23 [Fig. 18(b)] [45]. The LDCR retrieved VSM using the LSC function and *in situ* VSM at 10 cm are correlated with a coefficient of 0.69 [Fig. 18(c)].

The LDCR VSM data retrieved using the LSC and UCLSC functions over the period from June 14 to June 23 were compared with irrigation records for each land usage section (Table IV). In this comparison, the VSM data retrieved using the LSC function was averaged over each land usage section. From June 14 to June 20 it is noted that the wheat in the lateral area became wetter by 6.4%–10% while the corn and soybeans in the lateral area dried by 2%–6.5%. These changes

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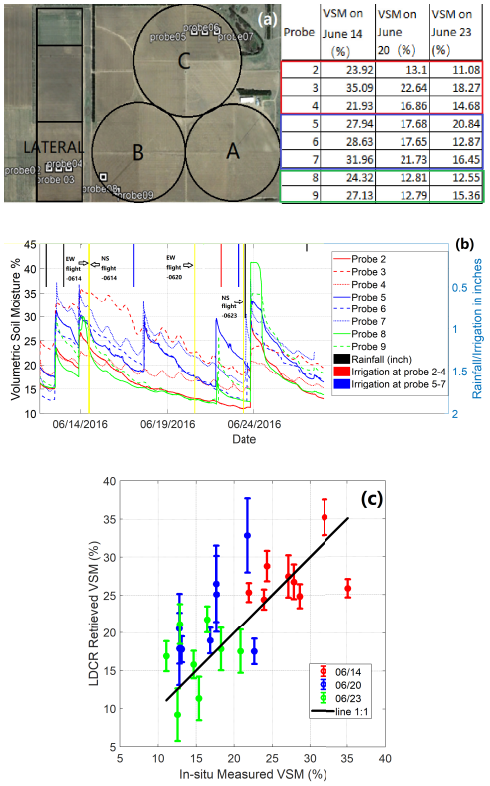

Probe	VSM on June 14 (%)	VSM on June 20 (%)	VSM on June 23 (%)
2	23.92	13.1	11.08
3	35.09	22.64	18.27
4	21.93	16.86	14.68
5	27.94	17.68	20.84
6	28.63	17.65	12.87
7	31.96	21.73	16.45
8	24.32	12.81	12.55
9	27.13	12.79	15.36

Fig. 18. *In situ* VSM data (a) during flights on June 14, 20, and 23, along with probe locations, (b) time history of *in situ* VSM, precipitation, and irrigation records, and (c) comparison between *in situ* and retrieved co-located LDCR VSM data.

are consistent with irrigation of the wheat by 1 in of water on June 17. It is also noted that the NE quarter of circle C dried by 0.9%–2.0% while the other 3 quarters dried by 1.9%–2.8%. These changes are consistent with irrigation of only the NE quarter of circle C by 0.5 in of water on June 17. We finally note that the NE quarter of circle A dried by 6.1%–6.4% while the other three quarters dried by 7.6%–8.6%. These changes are consistent with irrigation of only the NE quarter of circle A by 0.5 in of water on June 17, but also might be due to a smaller water demand by the grain sorghum relative to that of the corn.

From June 20 to June 23, it is noted that the soybeans in the lateral have approximately the same soil moisture while the wheat and corn dried by ∼8%. These changes are consistent with irrigation of the soybeans by 0.5 in of water on June 22. In circle B, the north half became wetter by 5.2%–9.5% while the south half dried by 0.7%–3.6%. These changes are consistent with irrigation of the north half by 0.5 in of water on June 21. In other cases reduction in averaged moisture is sometimes observed after an irrigation event, but it is possible that these reductions are caused by the water demand of the specific crops. The water demand for different crops will be compared against irrigation records and VSM changes in a future study.

D. VSM Estimation Error Analysis

The EW flight data on June 14 can be used for a nominal VSM estimation error analysis. Assuming all parameters are accurately known, VSM estimation error maps using various VSM spatial covariance functions were calculated and are compared in Fig. 19(a)–(c). It is noted that for VSM retrieval using the LSC function, the estimation errors at the boundaries between different land sections are increased by ∼1%–2% VSM compared with the estimation errors provided by a uniform spatial covariance function. These increases near boundaries are due effectively to a reduction of the data volume of the relevant set of observations at land section boundaries. For VSM retrieval using the UCLSC function, the estimation error has a uniform value within each managed land section. This estimation error is mostly below ∼0.2%. This low value is a result of the weighted averaging of all available observations over each of the 31 identified land usage sections. The number of spatially independent antenna temperature measurements within each land usage section varies from ∼15 to ∼250, thus resulting in a considerable increase in effective integration time relative to that of the LSC and the uniform spatial covariance methods. By combining of the EW and NS flight data, the estimation error is decreased as shown in Fig. 19(d). The minimum error is ∼3% beneath the intersection of the orthogonal flight lines, and maximum error is ∼5.5% between neighboring flight lines while the maximum estimation error between flight lines is ∼6.5%–7.5% with EW flight data only. The minimum VSM error is approximately equal to the value of sensor $\Delta T_{\mathrm{rms},s}$ propagated using a nominal 1 K/% T_A response to VSM.

The individual parameter uncertainty and the VSM estimation operator nonlinearity contributions to the overall VSM estimation error were determined for corn, wheat, and soybeans (Table V). For purpose of illustration, the error bias

TABLE IV
IRRIGATION AND LDCR VSM FOR VARIOUS IRF LAND SECTIONS

Sections (vegetation cover)	06/14 VSM ⟨LSC⟩	Irrigation	06/20 VSM ⟨LSC⟩	Irrigation	06/23 VSM ⟨LSC⟩
	UCLSC		UCLSC		UCLSC
Lateral (corn)	31.7%		29.8%		21.5%
	28.1%		25.5%		24.6%
Lateral (soybean)	24.8%		18.3%	0.5 inch on 06/22	17.6%
	24.9%		17.7%		20.9%
Lateral (wheat)	20.0%	1 inch on 06/17	26.4%		17.7%
	17.2%		27.2%		20.3%
Circle A NE (sorghum)	25.4%	0.5 inch on 06/17	19.3%	0.5 inch on 06/21	17.9%
	25.0%		18.6%		17.4%
Circle A (corn)	30.4%		21.8%	0.5 inch on 06/20	15.4%
	28.3%		20.7%		18.3%
Circle B north half (corn)	27.0%		15.9%	0.5 inch on 06/21	21.1%
	25.8%		14.3%		23.8%
Circle B south half (corn)	26.6%		20.0%		16.4%
	24.6%		18.1%		17.4%
Circle C NE (sugar beets)	30.3%	0.5 inch on 06/17	28.3%		23.2%
	30.2%		29.3%		26.2%
Circle C (corn)	25.5%		22.7%	0.5 inch on 06/20	14.3%
	23.6%		21.5%		16.1%
Non-irrigated land	20.2%		15.1%		10.3%
	20.0%		15.8%		11.5%

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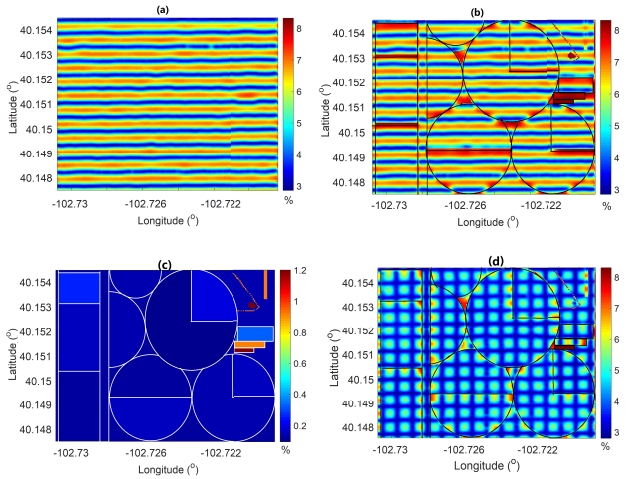

Fig. 19. LDCR VSM estimation error for June 14 using EW flight data only and (a) uniform spatial covariance function, (b) LSC function, and (c) UCLSC function, and (d) combing both EW and NS flight data.

TABLE V
VSM ESTIMATION ERROR DUE TO INTEGRATION ERROR, PARAMETER UNCERTAINTIES, AND ESTIMATION OPERATOR NONLINEARITY (ALL VSM ERROR VALUES IN PERCENTAGE ARE ROUNDED TO 2 DECIMAL PLACES)

Error Source	Corn		Wheat		Soybeans	
	Beneath Flight Line	Between Flight Line	Beneath Flight Line	Between Flight Line	Beneath Flight Line	Between Flight Line
$\Delta T_{RMS,s} = 2.58$ K	3.76%	6.66%	3.78%	6.73%	3.88%	6.77%
$\beta_{e_{W_v}} = 0.2$ kg/m^2	1.53%	0.74%	1.71%	0.70%	1.54%	0.29%
$\sigma_{e_{W_v}} = 0.2$ kg/m^2	0.00%	0.00%	0.01%	0.00%	0.00%	0.00%
$\beta_{e_{T_P}} = 5$ K	0.54%	0.21%	0.56%	0.19%	0.53%	0.18%
$\sigma_{e_{T_P}} = 5$ K	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
$\beta_{e_S} = 5\%$	0.08%	0.00%	0.08%	0.00%	0.07%	0.02%
$\sigma_{e_S} = 5\%$	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
$\beta_{e_C} = 5\%$	0.01%	0.00%	0.02%	0.00%	0.01%	0.00%
$\sigma_{e_C} = 5\%$	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
$\beta_{e_{b_h}} = 0.02$	0.42%	0.10%	0.33%	0.12%	0.53%	0.18%
$\beta_{e_{b_v}} = 0.02$	0.13%	0.00%	0.10%	0.00%	0.17%	0.02%
$\beta_{e_{h_R}} = 0.02$	0.10%	0.00%	0.10%	0.00%	0.10%	0.00%
$\beta_{e_\omega} = 0.01$	0.10%	0.00%	0.10%	0.00%	0.10%	0.00%
Nonlinearity	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Total	5.41%	7.11%	5.80%	7.33%	5.70%	7.28%

and variance of each parameter were set to be identical for all vegetation types. The VWC estimation bias and covariance used is 0.2 kg/m^2 [44], and the T_P bias error and variance used is 5 K [51]. Sand and clay bias percentage error and variance are set to be 5% [15]. For each vegetation type, estimation error beneath the flight lines and in between flights lines is compared. It is noted that among all parameters, the estimation bias in VWC W_v has the largest impact on the VSM estimation error, and T_P bias has the second largest impact. For the vegetation correction parameters, b_h has a larger impact than b_v which is consistent with higher attenuation at horizontal polarization from vegetation layers. Biases contribute larger errors than their associated variances for all geophysical measurements. The total estimation error is increased by these systematic uncertainties by ∼2% VSM beneath the flight lines and ∼0.5% between the flight lines.

IX. CONCLUSION

LDCR Rev A was built and integrated into the BST Tempest sUAS for high spatial resolution soil moisture mapping studies. Calibration was performed to determine the LDCR sensitivities and offset. The sensitivities and offset voltage thermal drift induced by the analog correlator chip self-heating was determined and compensated. Noise interference from the sUAS platform was observed and effectively mitigated by electromagnetic shielding. A total of 6 successful field experiments were performed in 2015 and 2016 at the Canton OK Soilscape and IRF field sties. The LDCR surface physical temperature T_P and antenna temperature T_A were measured on the sUAS sampling grid.

An LDCR high spatial resolution soil moisture mapping algorithm using the SVRT model and LMMSE full domain retrieval method was developed in this study. The SVRT-based VSM estimation method relates radiometric measurements at all sampling points to soil moisture at all user mapping points, and the VSM is retrieved based on the full-domain relations. Vegetation and surface roughness corrections are included in the SVRT model, and accounted for in the retrieval algorithm. A linearized observation operator for the LDCR antenna temperature observed on an arbitrary sUAS sampling grid due to VSM values on a user-defined mapping grid was determined. The LDCR VSM error covariance is determined using a physical LMMSE estimation method. The VSM estimation error was found to be primarily dependent on radiometer $\Delta T_{\mathrm{rms},s}$ and flight line separation. The vegetation correction is also critical to improve the LDCR VSM estimation accuracy. The retrieved LDCR soil moisture maps show good agreement with *in situ* VSM measurements and irrigation records, thus suggesting a viable new high spatial resolution airborne means of soil moisture mapping.

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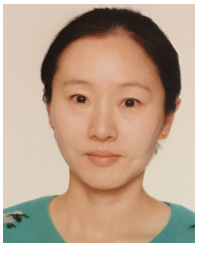

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